

A Comparative Study of Urban Greenery Trends in
Chennai and Shanghai: Analysis, Prediction, and
Sustainable Urban Planning Using NDVI and Machine
Learning



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1.Introduction

1.1 Urbanization and Environmental Trade-offs

The 21st century has seen unprecedented levels of urbanization. With cities becoming epicenters of economic activity and social mobility, millions are migrating to urban centers in search of better opportunities. However, this urban migration comes at a cost. Forests are cleared, wetlands are drained, and green spaces are replaced by concrete structures, highways, and industries. The resulting environmental trade-offs lead to issues such as:

Urban Heat Islands (UHIs)

One of the most prominent effects of reduced greenery in urban areas is the Urban Heat Island (UHI) phenomenon. As cities grow, natural vegetation is replaced with materials like asphalt, concrete, metal, and glass — all of which have high thermal mass and heat retention capabilities. These surfaces absorb solar radiation throughout the day and release it slowly at night, leading to consistently elevated temperatures in urban areas compared to their rural surroundings. This temperature difference can range from 2°C to as high as 7°C in some cities. The absence of vegetation, which normally helps regulate temperatures through shading and evapotranspiration, exacerbates this heat build-up. As a result, urban dwellers face increased discomfort during hot seasons, higher energy costs due to greater reliance on air conditioning, and greater health risks — especially for vulnerable populations like the elderly and those with preexisting medical conditions.

Air Pollution

Trees and green vegetation play a crucial role in maintaining urban air quality by acting as natural air filters. Their leaves and surfaces trap airborne pollutants such as particulate matter (PM2.5 and PM10), nitrogen dioxide, sulfur dioxide, and even heavy metals. They also contribute to oxygen production and carbon dioxide absorption. However, when green cover is reduced due to urban expansion, the city loses this vital ecological service. As vehicular emissions, industrial discharges, and construction dust rise, the lack of vegetation means more pollutants remain suspended in the air. This contributes directly to deteriorating respiratory health among urban populations, increases the prevalence of asthma and lung diseases, and reduces overall life expectancy, especially in densely populated and industrially active zones.

Flooding

Vegetated areas are naturally permeable, allowing rainwater to infiltrate the soil, replenish groundwater aquifers, and reduce surface runoff. When these green areas are replaced by impermeable surfaces such as roads, pavements, and buildings, rainwater is unable to soak into the ground. Instead, it rapidly accumulates and overwhelms stormwater drainage systems, leading to frequent urban flooding. Cities like Chennai have become increasingly prone to seasonal flooding due to this very pattern. The reduction in infiltration also contributes to long-term groundwater depletion, making cities more vulnerable to drought and water scarcity during dry months. The combination of flood risk and water insecurity presents a serious sustainability challenge for fast-growing cities.

Loss of Biodiversity

Urban green spaces such as forests, wetlands, and open parks are not only important for humans — they serve as habitats for a wide range of flora and fauna. These ecosystems support birds, insects (especially pollinators like bees and butterflies), amphibians, small mammals, and various plant species. The destruction of these spaces for infrastructure development leads to habitat fragmentation and loss, forcing many species to migrate or face extinction. This loss of biodiversity disrupts ecological balances, such as pollination and pest control, and contributes to the weakening of local food webs. Additionally, the disappearance of certain native species has cultural and ecological implications that are often irreversible.

Mental and Physical Health Decline

Access to nature and green environments is closely tied to the psychological and physiological well-being of urban populations. Numerous studies have shown that individuals who regularly interact with natural spaces experience lower stress levels, reduced anxiety, and improved mood. Greenery also promotes physical activity, encourages social interaction, and reduces the effects of environmental stressors like noise and pollution. In contrast, the absence of green spaces in urban settings can lead to feelings of isolation, increased risk of depression, and higher prevalence of non-communicable diseases such as obesity and cardiovascular conditions. Vulnerable groups, including children and the elderly, are especially affected. Thus, maintaining and expanding urban green spaces is not merely an aesthetic or environmental concern, but a fundamental aspect of public health strategy.

1.2 Monitoring Vegetation with NDVI

The Normalized Difference Vegetation Index (NDVI) has become one of the most widely accepted and utilized tools for monitoring vegetation health and distribution across the globe. Developed through advancements in satellite-based remote sensing, NDVI provides a simple yet powerful way to quantify plant activity and biomass using spectral reflectance data. It is extensively used in disciplines such as agriculture, climate science, forestry, and urban environmental monitoring due to its reliability, consistency, and accessibility.

At the core of NDVI's calculation is the relationship between two key spectral bands: near-infrared (NIR) and red (RED). Healthy green vegetation absorbs most of the visible red light (due to chlorophyll) and reflects a large portion of the near-infrared light. In contrast, barren surfaces, urban areas, or water bodies reflect both red and NIR in lower, less distinctive proportions. The formula for NDVI is expressed as:

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$

Where:

- **NIR (Near-Infrared)** is strongly reflected by healthy vegetation
- **RED** is absorbed by chlorophyll in plants

Interpretation:

- **NDVI > 0.6:** Dense green vegetation (forests, healthy crops)
- **NDVI 0.2 – 0.6:** Sparse vegetation (grasslands, shrubs)
- **NDVI < 0.2:** Barren land, urban areas, or water

In this equation, NIR refers to the near-infrared light reflectance, while RED corresponds to the red light reflectance captured by satellite sensors. The result of this computation is a normalized index ranging from -1 to +1. Values close to +1 indicate dense, healthy vegetation, while values near 0 suggest minimal or sparse vegetation. Negative values are generally associated with water bodies, clouds, snow, or artificial structures such as concrete and asphalt.

NDVI values are typically interpreted using the following ranges: values above 0.6 represent areas with dense, lush vegetation such as tropical forests, healthy agricultural zones, or well-maintained urban green parks. Values between 0.2 and 0.6 reflect moderate vegetation cover, including grasslands, shrubs, or partially

vegetated landscapes. Anything below 0.2 indicates sparse or no vegetation, which can include deserts, urbanized land, rocky terrains, or regions under severe ecological degradation.

One of NDVI's most powerful features is its ability to track changes in vegetative cover over time. By comparing NDVI imagery across different time periods, researchers can identify areas experiencing vegetation loss due to deforestation, infrastructure development, drought, or natural disasters. Conversely, it can also detect areas showing vegetation recovery through afforestation, agricultural interventions, or urban greening programs. When this data is analyzed over extended timeframes, such as decades, it reveals trends that help policymakers, urban planners, and environmental scientists understand the long-term ecological trajectory of a region.

In predictive modeling, NDVI is also critical. By integrating NDVI data into machine learning models alongside other socio-economic and environmental variables, researchers can forecast how vegetation patterns are likely to evolve under various development or climate scenarios. This predictive capacity makes NDVI not only a monitoring tool but also a proactive planning instrument that informs policy decisions and conservation strategies.

In the context of this study, NDVI has been used to assess the greenery in Chennai and Shanghai over the past two decades. Historical NDVI datasets were compiled and preprocessed to reveal trends in vegetation change, while predictive NDVI values for the next 20 years were generated using machine learning techniques. This NDVI-centered approach forms the scientific foundation for the visualizations, maps, and comparative assessments featured in the report.

1.3 Defining the Problem: Urban Green Decline vs Growth

This section introduces a comparative perspective on the divergent paths of vegetation change taken by two major global cities — Chennai, India, and Shanghai, China. While both cities have undergone rapid urbanization, their environmental strategies have produced opposite outcomes: one marked by degradation, the other by regeneration. Understanding these patterns not only sheds light on how policy and planning influence urban greenery but also reveals opportunities for adaptive change in regions facing ecological stress.

Chennai, India

High rates of unregulated urban sprawl

Chennai has experienced extensive horizontal expansion in the past two decades, often without formal regulation or master planning. Residential and commercial construction projects have spilled beyond designated urban zones, encroaching upon wetlands, forests, agricultural lands, and other green spaces. This sprawl has occurred faster than civic infrastructure and environmental policies could adapt, leading to the unchecked destruction of natural habitats and a steep decline in tree cover and open spaces.

Increased surface temperature and flooding incidents

As vegetation-rich areas were replaced with impervious surfaces like asphalt and concrete, the city's ability to regulate temperature naturally diminished. This transformation has amplified the urban heat island (UHI) effect, making Chennai significantly warmer than surrounding rural areas. Simultaneously, the reduction of permeable ground has led to higher surface runoff during monsoons. With the disappearance of natural drainage systems and tree roots that absorb water, flooding events have become more frequent and severe.

Decline in urban tree cover

Tree cover in Chennai has been shrinking at an alarming rate. Trees, which play a crucial role in providing shade, cooling the environment, absorbing pollutants, and preventing soil erosion, are often the first casualties of road expansion and real estate development. The absence of robust urban forestry programs has resulted in the systematic removal of roadside and parkland trees, with limited replanting or compensation.

Lack of consistent policy for green infrastructure

Although several municipal programs have attempted to address urban greening,

Chennai lacks a unified, long-term strategy for environmental sustainability. Green policies are often reactive — implemented after flooding or public outcry — rather than preventive. Coordination between government departments remains weak, leading to fragmented efforts that fail to create lasting impact. There is also a gap in enforcing tree-cutting regulations and mandating green buffers in construction projects.

Rising population density without proportional increase in green zones

Chennai's growing population has placed enormous pressure on its limited land resources. New housing, roads, and industries have proliferated without a corresponding investment in public parks, community gardens, or green rooftops. As a result, per capita access to green space has diminished significantly. This lack of nature access negatively impacts both mental and physical well-being, particularly in densely packed low-income neighborhoods where open space is scarce.

Shanghai, China

Rapid urbanization with a parallel green growth strategy

Unlike Chennai, Shanghai's urban development has been guided by a parallel investment in environmental sustainability. As skyscrapers, highways, and tech hubs emerged, city planners embedded green corridors, forest belts, and ecological zones into the blueprint. This foresight ensured that Shanghai's urban footprint expanded without sacrificing ecological resilience. The city adopted a "sponge city" model — urban design aimed at natural water retention and green integration — ensuring sustainable coexistence of buildings and vegetation.

Aggressive tree planting campaigns

Shanghai has institutionalized large-scale, citywide tree-planting initiatives aimed at restoring canopy cover and enhancing biodiversity. Annual planting targets are set and met with the involvement of local governments, schools, and community groups. These campaigns focus not only on quantity but also on planting native species, ensuring long-term growth and ecological balance. As a result, Shanghai has significantly expanded its urban forest cover over the years.

Promotion of rooftop gardens and green belts

To counterbalance vertical construction, the city has encouraged the greening of rooftops, balconies, and walls. Building codes incentivize or mandate the inclusion of green infrastructure in both residential and commercial buildings. Rooftop gardens not only reduce building heat but also provide habitat for pollinators and improve air

quality. Meanwhile, green belts have been established around industrial zones, highways, and urban peripheries, reducing pollution and creating buffer zones.

Enforcement of land zoning laws for urban forest conservation

Land use in Shanghai is tightly controlled through clear zoning regulations. Certain areas are legally protected for ecological preservation and are not available for commercial development. This enforcement ensures that green spaces are retained even in the face of economic pressure. Violators face steep fines and project delays, which discourages illegal deforestation and fosters a culture of compliance with environmental law.

Large-scale community and governmental involvement

Shanghai's green success story is also rooted in its participatory approach. Government-backed environmental awareness campaigns engage the public through festivals, workshops, and educational programs. Schools include sustainability modules in their curriculum. Volunteer groups regularly contribute to cleanup drives, garden maintenance, and citizen science projects. The government, in turn, supports these efforts through funding, infrastructure, and recognition — reinforcing a positive feedback loop between state and society.

This comparison illustrates how two cities, facing similar challenges, have diverged due to their approach toward green infrastructure. The evidence from Shanghai provides practical examples of how proactive planning, legal frameworks, and civic participation can reverse ecological decline — lessons that can be potentially adapted and localized for Chennai's context.

1.4 Study Objectives

This study is designed to provide a comprehensive, data-driven evaluation of vegetation dynamics in two of Asia's major urban centers — Chennai, India, and Shanghai, China — with a focus on understanding their contrasting trajectories in urban greenery over the past two decades, and predicting their potential futures. The objectives are structured to guide the analysis from historical observation to policy recommendations, ensuring a holistic view of both ecological trends and practical interventions.

Long-Term Greenery Analysis (2005–2024)

The foundation of this study lies in a thorough examination of vegetation changes from 2005 to 2024, using NDVI (Normalized Difference Vegetation Index) data derived from satellite imagery. This 20-year timespan provides a sufficiently broad window to capture meaningful trends and seasonal variabilities in green cover. NDVI values, calculated from remote sensing datasets, serve as proxies for vegetation health and density, enabling a consistent and objective measure of change over time.

The temporal NDVI data is correlated with key events such as major urban development projects, policy shifts, climatic anomalies (like droughts or heavy monsoons), and economic transitions. In Chennai, for example, periods of rapid infrastructure expansion are cross-referenced with noticeable dips in NDVI values. In Shanghai, efforts like the "Green City" program are similarly overlaid with NDVI upticks, revealing the success of environmental governance. This layered temporal approach enables a deeper understanding of cause-and-effect relationships between urban growth and ecological balance.

Factor Analysis: Understanding Drivers Behind Green Cover Trends

To uncover the underlying causes of greenery increase or decline, a multi-variable analytical approach is employed. Several interrelated datasets are integrated to assess how human and environmental variables have influenced vegetation patterns:

- **Urban Land Use Change:** Changes in land classification from agricultural or open land to built-up zones are tracked using GIS and satellite-derived land cover datasets. This helps quantify how urban sprawl affects green patches.
- **Population Growth:** Census data and population density maps are used to understand demographic pressures. Increasing population often leads to housing demands that encroach upon green belts and forest edges.

- **Climate Variables:** Longitudinal records of temperature and rainfall are analyzed to assess climatic influence on vegetation patterns. For instance, prolonged droughts or heatwaves correlate with NDVI dips, while good monsoon years support vegetation spikes.
- **Air Pollution:** Concentrations of pollutants such as NO₂, PM2.5, and SO₂ are examined. Green cover naturally mitigates air pollution, but when vegetation declines, pollution can rise, creating a feedback loop that inhibits regrowth.

Each of these variables is statistically correlated with NDVI values using scatter plots, correlation matrices, and regression techniques, revealing the strength and direction of influence each factor holds.

Future Greenery Prediction (2025–2044)

After understanding historical patterns, the study employs predictive modeling to forecast vegetation trends for the next 20 years, i.e., from 2025 to 2044. A combination of machine learning models is used to make these forecasts more robust and credible:

- Linear Regression and Polynomial Regression provide a simple but interpretable baseline model.
- Time Series Forecasting (using ARIMA or Prophet models) accounts for temporal dependencies and seasonality in NDVI data.
- LSTM (Long Short-Term Memory), a deep learning model specialized in sequential data, is used to capture non-linear trends and long-term dependencies in NDVI time series.

Each model is validated using cross-validation, and their accuracy is assessed using error metrics such as Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE). The most accurate model is then selected to forecast NDVI values for each year from 2025 to 2044, forming the basis for visualization and policy formulation.

Visualizing Green Futures with Satellite Imagery

A key innovation in this study lies in translating abstract NDVI forecasts into realistic, image-based projections of future green cover. Using high-resolution base satellite maps of Chennai and Shanghai, green pixels are dynamically modified to reflect the predicted NDVI values for each year. This method not only makes future greenery trends more visually intuitive, but also enhances their persuasive power when

communicating with non-technical stakeholders such as city planners, community leaders, and the general public.

For Chennai, the visual maps depict a gradual decline in greenery unless interventions are implemented, whereas Shanghai's maps illustrate how proactive planning leads to richer and more widespread vegetation coverage. These images act as powerful tools for storytelling, raising awareness and driving policy support.

City Comparison and Policy Transferability

Finally, the study culminates in a comparative analysis between the two cities. By identifying critical policy and infrastructure decisions that helped Shanghai reverse or mitigate the environmental costs of urbanization, we explore how similar approaches could be adapted for Chennai. The comparison includes:

- Analyzing turning points in Shanghai's green strategy (e.g., zoning reforms, urban forest policies)
- Evaluating the feasibility of importing those strategies into Chennai's governance, socio-economic, and climatic context
- Proposing localized adaptations of Shanghai's successes — including recommendations for community-based programs, regulatory changes, and budget allocations

Through this comparative lens, the study not only draws contrasts but also charts a roadmap for Chennai's ecological restoration, with the potential for broader application across other urban centers in the Global South.

1.5 Global Significance and Relevance

While the primary focus of this study is the detailed analysis of urban greenery trends in Chennai and Shanghai, its implications extend far beyond the borders of these two cities. The findings, methodologies, and policy recommendations derived from this research contribute directly to the global discourse on urban sustainability, environmental resilience, and the future of ecological planning in an era of accelerating climate change. As cities around the world become increasingly urbanized, the role of urban greenery becomes critical—not only for environmental balance, but also for public health, climate mitigation, and socio-economic wellbeing.

Alignment with Global Frameworks: The UN Sustainable Development Goals

This study supports and advances several United Nations Sustainable Development Goals (SDGs), highlighting its international relevance:

SDG 11 – Sustainable Cities and Communities

This goal emphasizes the creation of inclusive, safe, resilient, and sustainable urban spaces. By investigating urban green cover as a measurable and essential component of livable cities, the study provides actionable data and projections that can help policymakers design better, greener urban environments. The contrasting case studies of Chennai and Shanghai serve as examples of challenges and solutions in achieving SDG 11.

SDG 13 – Climate Action

Urban greenery plays a pivotal role in climate mitigation and adaptation. Trees and vegetation not only sequester carbon dioxide but also reduce urban temperatures, regulate water cycles, and prevent soil erosion. This study's predictive modeling of NDVI trends offers insights into the long-term climate resilience of urban ecosystems, aligning closely with the targets under SDG 13. By showcasing the future implications of current trends, the research calls attention to the urgency of climate-informed urban planning.

SDG 15 – Life on Land

This goal advocates for the protection, restoration, and sustainable use of terrestrial ecosystems. In the context of expanding cities, preserving green zones and promoting reforestation within urban boundaries supports biodiversity, improves air and soil quality, and maintains ecological balance. The study's focus on green loss and regeneration in heavily populated areas like Chennai, alongside success stories like Shanghai, presents a valuable contribution to the implementation of SDG 15 at the urban level.

Contextual Relevance in Today's Climate Challenges

The study holds heightened relevance in today's environmental context. The world is witnessing an increase in urban heatwaves, water scarcity, air pollution, and extreme weather events, all of which are exacerbated by rapid urbanization and the depletion of green infrastructure. In this setting:

- Urban heat islands (UHIs) are intensifying, raising health risks and energy demands in cities

- Flooding incidents are becoming more common due to reduced green absorption surfaces
- Mental and physical health outcomes are declining with the loss of accessible green spaces

In light of these issues, green urbanism is not an aesthetic luxury—it is an ecological and humanitarian necessity. This research, therefore, responds to a growing global imperative to reintroduce nature into the built environment through strategic, data-backed interventions.

Influencing Stakeholders Through a Visual, Evidence-Based Approach

One of the core strengths of this study lies in its visual storytelling, which makes complex environmental data accessible to a wide range of audiences. The use of real satellite imagery, predictive NDVI modeling, and future landscape simulations ensures that the research does not remain confined to academic circles, but instead has the power to inform and influence a broad set of stakeholders:

Urban Planners

For professionals involved in city design and infrastructure development, the study offers critical data on how different planning approaches (like those seen in Shanghai) influence long-term ecological outcomes. The visual maps and trend analyses can guide zoning laws, park allocation, rooftop greening programs, and building codes that incorporate green buffers.

Environmental Policymakers

Government and institutional decision-makers can leverage the report's findings to draft or amend green policies, allocate environmental budgets, and incentivize community-based greening initiatives. The comparative approach shows how policy success in one region may be adapted to others with contextual modifications.

Citizen Groups and Environmental Advocates

By translating remote sensing data into clear narratives, the study empowers civil society groups and local organizations to campaign more effectively for green spaces, raise public awareness, and hold urban authorities accountable for environmental degradation.

Climate Researchers

The methodological rigor, particularly in the use of machine learning for future vegetation prediction, provides a framework for climate scientists and data researchers to explore NDVI-linked patterns in other urban areas. It contributes to the growing body of literature around AI-assisted environmental forecasting.

City Administrators and Government Bodies

Municipal officials responsible for day-to-day governance can use the study to balance growth with ecological sustainability. It highlights how long-term planning and short-term interventions can work together to create greener, more livable urban spaces.

2. Study Areas: Context and Significance

2.1 Chennai

2.1.1 Overview of Urban Expansion

Chennai, the capital of Tamil Nadu and a major metropolitan hub in South India, has undergone rapid urbanization over the past two decades. Its economic transformation—particularly in the sectors of information technology, real estate, and manufacturing—has driven population growth and the subsequent expansion of city boundaries. Previously rural and semi-urban areas have now become part of Greater Chennai. However, this expansion has largely been unplanned, with many new developments sprouting up in ecologically sensitive zones, often without adequate environmental impact assessments or green space allocation. Traditional land-use patterns have given way to high-density urban settlements, placing enormous strain on the city's natural resources and infrastructure.

2.1.2 Environmental Challenges (Heat, Flooding, Pollution)

The consequences of unregulated urban expansion are evident in the form of severe environmental stress. The Urban Heat Island (UHI) effect has become more pronounced, with average land surface temperatures rising each year due to widespread use of heat-absorbing materials like concrete, metal, and asphalt. Simultaneously, Chennai's vulnerability to flooding has escalated due to the destruction of natural drainage systems such as lakes, marshes, and wetlands. Iconic water bodies like the Pallikaranai Marsh have shrunk drastically, impacting both flood mitigation and biodiversity. Air pollution has also worsened, partly due to a rise in vehicle emissions and the declining number of trees which would otherwise help filter pollutants. Many parts of the city now regularly record PM_{2.5} and NO₂ levels above safe limits, contributing to public health risks.

2.1.3 Current Green Space Distribution

Chennai's current green space distribution is patchy and imbalanced. Legacy green zones like Guindy National Park, Theosophical Society gardens, and IIT Madras forest campus remain as isolated pockets of biodiversity. However, newly urbanized neighborhoods often lack dedicated parks, tree cover, or green buffers. Urban planning in newer developments typically prioritizes concrete infrastructure over ecological integration. According to recent estimates, Chennai falls short of the WHO recommendation of 9 square meters of green space per person. Moreover, encroachments on public lands and poor maintenance of existing parks further exacerbate the city's green deficit. This uneven distribution also contributes to

environmental injustice, where access to nature becomes a privilege limited to wealthier neighbours.

2.2 Shanghai

2.2.1 Urban Planning Evolution

Shanghai, China's largest city and a global financial center, has experienced explosive urban development since the 1990s. However, unlike many rapidly urbanizing regions, Shanghai's growth has been largely governed by robust master planning. The Shanghai Urban Master Plan (2017–2035), among earlier versions, emphasized not only economic development but also environmental quality and sustainability. This foresight allowed the city to avoid many of the ecological pitfalls commonly associated with urbanization. Transit-oriented development, vertical expansion to preserve ground-level land, and the integration of green belts into city zoning all reflect a comprehensive approach to urban growth. Shanghai's model blends high-density urbanism with ecological sensibility, ensuring that expansion does not come at the cost of environmental degradation.

2.2.2 Government-led Afforestation Drives

Shanghai's transformation into a greener city has been driven by sustained governmental intervention. The city launched a series of afforestation campaigns aimed at increasing urban forest cover and reversing environmental damage. These initiatives included converting vacant or underutilized land into green zones, planting native tree species, and encouraging afforestation in peri-urban areas. By 2020, Shanghai had created over 400 "pocket parks" and significantly expanded urban forest coverage. The city also established ecological red lines—protected zones that limit development and preserve biodiversity. Many of these greening efforts were not isolated events but part of long-term strategic programs with measurable targets, citizen participation, and strong institutional backing.

2.2.3 Environmental Success Stories

The results of Shanghai's green transformation are both visible and measurable. The Urban Forest Plan helped increase the city's forest coverage to over 16% by the mid-2020s, significantly improving air quality and microclimates in densely populated districts. The revival of the Suzhou Creek is another key milestone, where a once heavily polluted waterway was restored and lined with green walkways, public parks, and eco-friendly infrastructure. Additionally, the city has promoted rooftop gardens, vertical greening of buildings, and green roofs in commercial spaces. These measures have contributed to reduced ambient temperatures, enhanced biodiversity, and improved mental well-being for residents. Shanghai now serves as a model for

integrating nature into modern urban environments, offering key lessons for cities facing ecological degradation.

2.3 Comparative Insight: Why These Cities Matter Together

Chennai and Shanghai, while geographically and culturally distinct, offer a compelling contrast in urban environmental trajectories. Both cities have undergone rapid urbanization over the past two decades, fueled by population growth, industrial expansion, and infrastructural development. However, their approaches to managing this urban growth—especially concerning green space preservation and environmental health—have diverged significantly.

Chennai's urban expansion has been largely reactive, with limited forward-looking environmental planning. The city's green cover has steadily declined due to unchecked construction, poor zoning enforcement, and minimal investment in ecological buffers. As a result, Chennai is facing mounting environmental challenges such as recurring floods, urban heat islands, and deteriorating air quality. Green spaces, though historically abundant, have been marginalized in favor of built-up infrastructure, with few recent efforts to reclaim or restore them.

Shanghai, on the other hand, represents a model of proactive ecological urbanism. Its emphasis on environmental balance, embedded within city planning frameworks, has allowed for the integration of green strategies alongside economic growth. The expansion of forest belts, vertical gardens, and eco-corridors within urban spaces illustrates a deliberate and sustained effort to preserve nature even amidst large-scale development. Government policies, civic engagement, and well-funded greening initiatives have ensured that Shanghai's urban environment remains resilient and livable.

Studying these two cities together provides valuable insight into the long-term outcomes of differing urban development philosophies. It allows for an empirical comparison of how governance, public policy, and societal involvement can influence environmental well-being. More importantly, it raises a critical question: can lessons from a green-forward city like Shanghai be adapted to reverse or mitigate the ecological degradation seen in cities like Chennai? This comparison not only highlights areas of concern but also points toward actionable pathways for sustainable urban transformation.

3. Data Collection & Sources

3.1 NDVI Data

Source (e.g., MODIS, Landsat, Sentinel)

The NDVI data used in this study was sourced primarily from two globally recognized Earth observation programs: NASA's MODIS (Moderate Resolution Imaging Spectroradiometer) and the European Space Agency's Sentinel missions. MODIS provides continuous, long-term NDVI datasets with a wide global footprint, making it ideal for observing vegetation changes across decades. Landsat archives were also referred to for historical imagery with moderate resolution, especially for early years. More recently, Sentinel-2 data, with high spatial resolution and multispectral capabilities, was used to enhance the clarity and detail of vegetation observations. The selection of these sources was based on their scientific reliability, accessibility, and consistent temporal coverage.

Temporal resolution

Temporal resolution refers to how frequently satellite data is captured for a given location. MODIS offers NDVI data at a 16-day interval, allowing the creation of near-monthly composites that help reduce cloud interference and capture seasonal changes. Landsat has a revisit time of 16 days per satellite, while Sentinel-2 has a 5-day revisit period with its twin satellites. This high revisit frequency, especially from Sentinel-2, allowed the study to fill temporal gaps and ensure data continuity, particularly when monitoring rapidly changing urban environments. The combination of high temporal frequency and multi-source validation ensured robustness in capturing vegetation trends year by year.

Preprocessing steps

Before any analysis, raw satellite NDVI data underwent several preprocessing steps to improve quality and consistency. Cloud masking was applied using built-in quality flags provided by MODIS and Sentinel datasets, ensuring that only cloud-free pixels contributed to the NDVI composites. Next, geometric corrections were performed to align different satellite datasets spatially, followed by resampling to a uniform spatial resolution. Temporal aggregation was then used to create annual mean NDVI layers by averaging monthly values, reducing noise from short-term fluctuations. Finally, NDVI values were normalized to standard scales to ensure comparability between sensors and across years, which is critical when merging data from multiple satellite platforms.

3.2 Contributing Factors

3.2.1 Population Growth (Census, UN data)

Population growth is a fundamental driver of urban landscape transformation, especially in rapidly developing cities like Chennai and Shanghai. For this study, population statistics were drawn from official national census reports as well as international sources such as the United Nations World Urbanization Prospects. Chennai has witnessed a steep population rise from under 5 million in the early 2000s to over 8 million by 2024. This expansion has not been matched with proportional green infrastructure development. Rising housing demands and infrastructural expansion have consumed once-vegetated land parcels. The increased population density has led to the conversion of natural and semi-natural areas into residential and commercial blocks, thereby accelerating the loss of tree cover. On the other hand, Shanghai's population grew at a similarly rapid pace, but the city anticipated these changes by mandating green spaces in all new developments. Zoning laws and vertical city planning helped Shanghai limit horizontal sprawl while preserving key vegetative belts.

3.2.2 Urban Land Use (Satellite Classification)

Land use change plays a pivotal role in the health and distribution of urban greenery. In this research, land use patterns were examined using satellite-based land cover classification techniques, including supervised classification algorithms applied through tools like QGIS and Google Earth Engine. These methods allowed differentiation between vegetative areas, water bodies, built-up infrastructure, and barren lands. Historical satellite imagery revealed that Chennai has experienced a marked conversion of green zones—such as marshlands, parks, and peri-urban forests—into built-up areas. This transformation is especially pronounced along highways, coastal zones, and newly industrialized corridors. The haphazard expansion has fragmented green patches, reducing ecosystem connectivity and making the remaining vegetation more vulnerable. Shanghai's land use change, though also aggressive, was guided by long-term planning strategies. The city reserved spaces for urban forests and ecological corridors as part of its expansion, helping mitigate some of the usual ecological costs of urbanization.

3.2.3 Pollution Index (Government/Environmental Databases)

Air quality directly affects and is affected by vegetation cover. This study incorporated pollution metrics from national environmental monitoring agencies, real-time open platforms like OpenAQ, and long-term datasets from the World Health Organization. In Chennai, high levels of particulate matter (PM_{2.5} and PM₁₀),

nitrogen dioxide (NO₂), and sulfur dioxide (SO₂) have been recorded consistently, especially in industrial zones and high-traffic areas. The deterioration in air quality correlates strongly with the decline in tree cover, which otherwise would help in pollutant absorption and reduction. Fewer trees mean lower filtration capacity, leading to a compounding effect on air quality. Vegetation not only improves air quality but also shields human health. In Shanghai, despite high industrial activity, air quality has seen gradual improvement in certain zones thanks to green buffer zones and afforestation campaigns. Trees and shrubs have been planted near emission-heavy areas to act as living barriers, showcasing a strategic integration of pollution control and urban greening.

3.2.4 Rainfall (Climatic Data)

Rainfall patterns critically influence vegetation health, especially in urban environments where natural water flow is often disrupted. Data from the India Meteorological Department (IMD) and China Meteorological Administration, along with international datasets like WorldClim, were analyzed to trace annual and seasonal rainfall trends. Chennai experiences a monsoon-dependent climate, with most of its annual rainfall occurring between October and December. However, erratic monsoons and increasing dry spells over the past two decades have led to water stress in plants, especially in non-irrigated public spaces. The shrinking of wetlands and surface water bodies has further compounded the issue. Reduced rainfall weakens the root systems of trees and hampers new sapling growth. In contrast, Shanghai benefits from a more evenly distributed rainfall pattern throughout the year. The presence of integrated stormwater management systems and artificial irrigation in parks has helped Shanghai maintain vegetation health even during drier seasons.

3.2.5 Temperature (Climatic Data)

Urban temperature trends were studied using both satellite-derived land surface temperature (LST) datasets and ground-based weather station reports. Chennai has experienced a sharp rise in urban heat island (UHI) effects, particularly in central and high-density zones. Daytime summer temperatures have been observed to exceed 42°C in some years, significantly higher than surrounding rural areas. This increase in temperature is linked to the reduction in green zones, as vegetation plays a key role in evapotranspiration and natural cooling. High surface temperatures also contribute to soil dehydration, which impedes the survival of young vegetation and saplings. Shanghai has similarly experienced warming, but proactive cooling measures—such as green roofs, tree-lined streets, and water-integrated landscaping—have helped in

local heat regulation. Their focus on combining greenery with infrastructure (eco-bridges, vertical gardens, reflective surfaces) has allowed the city to manage rising temperatures more effectively than Chennai.

3.3 Satellite Map Imagery

Source of Base Images

For the purposes of map-based visualization, high-resolution satellite imagery was sourced from the Copernicus Sentinel Program, managed by the European Space Agency (ESA). The Sentinel-2 satellites, in particular, offer multispectral imaging at a spatial resolution of 10 to 60 meters, making them highly suitable for urban vegetation analysis.

Copernicus provides open-access satellite data that includes Near-Infrared (NIR) and Red bands essential for computing the Normalized Difference Vegetation Index (NDVI). The program's robust temporal coverage, with frequent revisits over the same geographic area, allowed consistent image capture for both historical and predictive visualization.

Images were downloaded and preprocessed using tools such as SNAP (Sentinel Application Platform) and QGIS, ensuring georeferencing consistency and noise reduction. These processed maps were then used as the foundation for overlaying NDVI predictions to visually demonstrate vegetation decline or growth in the selected cities.

By using Copernicus data, the study ensures both scientific reliability and transparency, as the platform adheres to international standards for earth observation.

4. NDVI-Based Historical Trend Analysis (2005–2024)

4.1 Chennai: Decline Pattern Visualization

The city of Chennai has undergone rapid and largely unplanned urban expansion over the past two decades. NDVI analysis for the years 2005 to 2024 shows a steady and visible decline in overall vegetation cover, with particularly sharp drops observed around periods of intensified infrastructure development, such as highway construction, metro expansion, and the proliferation of high-rise housing in peripheral zones. Green patches within the core city, especially in South Chennai and central business districts, have shrunk due to increased conversion of open land into commercial and residential complexes.

These changes were visualized through temporal NDVI heatmaps, which show the transition of once-green areas into low-vegetation or barren land. The trend line generated from NDVI values over this period exhibits a consistently downward slope, highlighting a major concern for future sustainability.

4.2 Chennai: Correlation with Urbanization, Rainfall, and Population

The declining NDVI trend in Chennai aligns strongly with several interlinked factors. Firstly, urbanization rates have surged, particularly in the post-2010 period, leading to extensive land use change. Remote sensing-based urban classification data confirms that open green lands and agricultural fringes have been transformed into built-up environments.

Simultaneously, there has been a noticeable reduction in annual rainfall variability, which, when overlaid with NDVI data, reveals a close relationship between vegetation loss and decreased water availability. Additionally, population data from census records shows a steep growth in population density, especially in core zones, without a proportional increase in green space allocation. Together, these factors have contributed significantly to the observed decline in Chennai's urban greenery.

4.3 Shanghai: Growth Pattern Visualization

In stark contrast to Chennai, Shanghai's NDVI data from 2005 to 2024 indicates a clear and gradual increase in vegetation density. The NDVI curve displays an upward trend, reflecting deliberate city-wide efforts to enhance green infrastructure. Visualization of this trend through heatmaps and overlaid NDVI images reveals expanding green belts around the city periphery, new forest zones, and densely planted areas along riverbanks and roadways.

Notably, rooftop gardens and vertical greenery initiatives have played a major role in increasing vegetation in densely populated districts where ground-level green space was limited. These patterns point toward a successful, top-down approach to urban greening that has transformed Shanghai’s environmental landscape over time.

4.4 Shanghai: Correlation with Investment in Green Projects

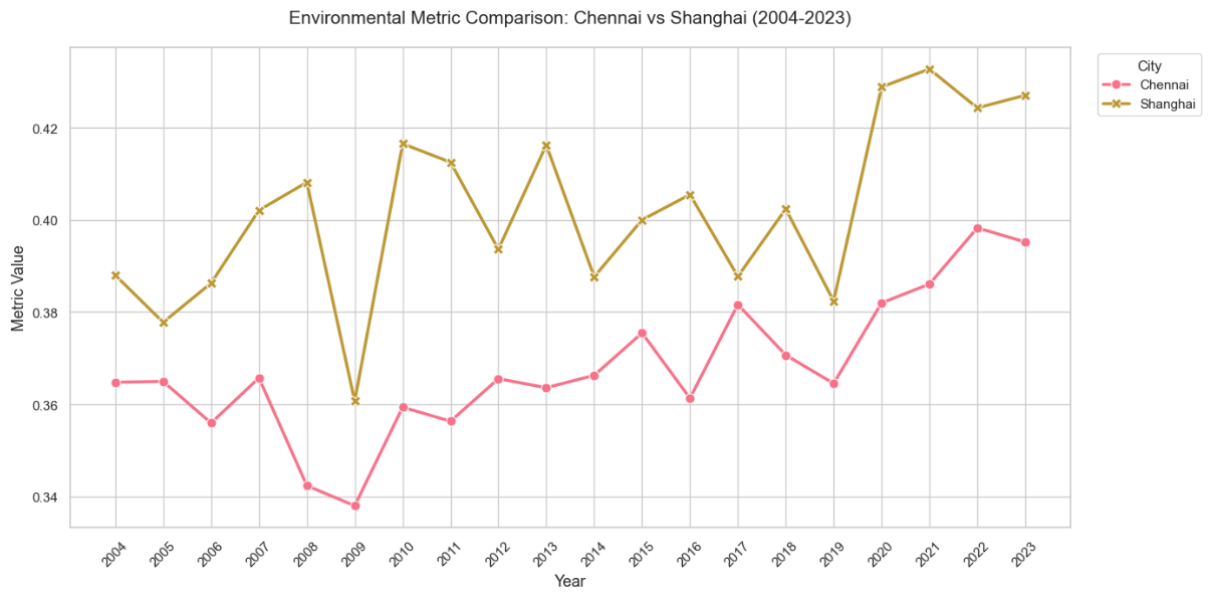
Shanghai’s growth in NDVI aligns strongly with large-scale governmental initiatives aimed at ecological restoration and urban beautification. Policy decisions such as the “Green for People” campaign, coupled with mandatory greening requirements for all new urban projects, have significantly influenced vegetation growth.

Data gathered from environmental planning reports and municipal budget documents indicate sustained investments in afforestation, ecological parks, and sustainable landscaping over the years. Additionally, public-private partnerships in green space development have played a major role in ensuring that even commercial and industrial developments incorporate green components.

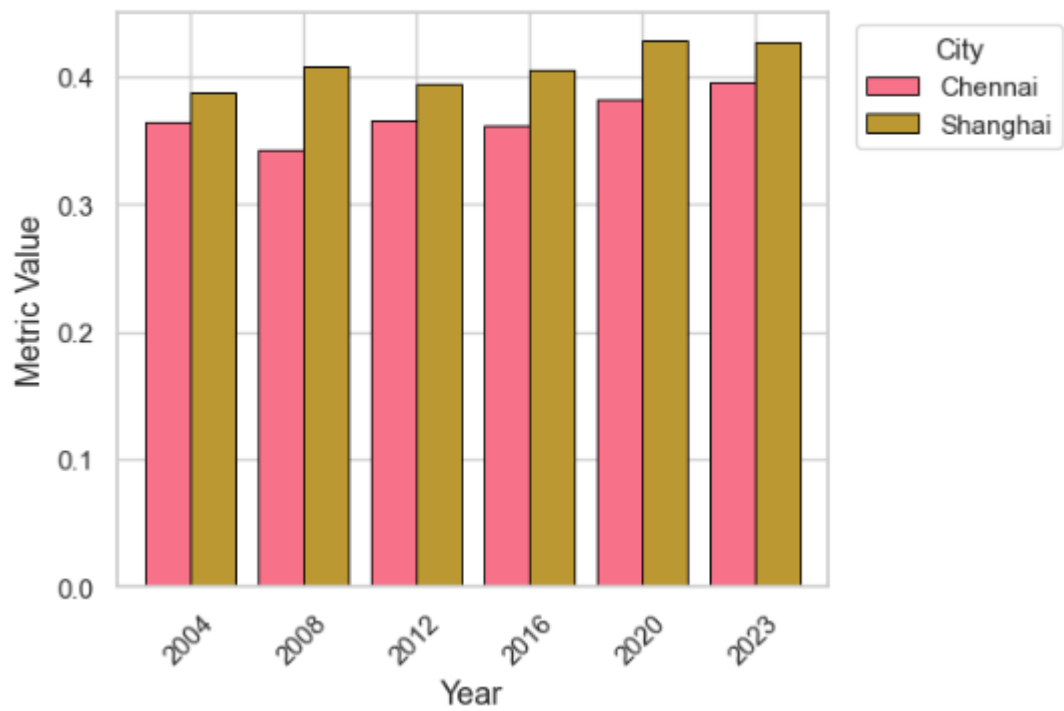
4.5 Visuals Supporting Historical Trends

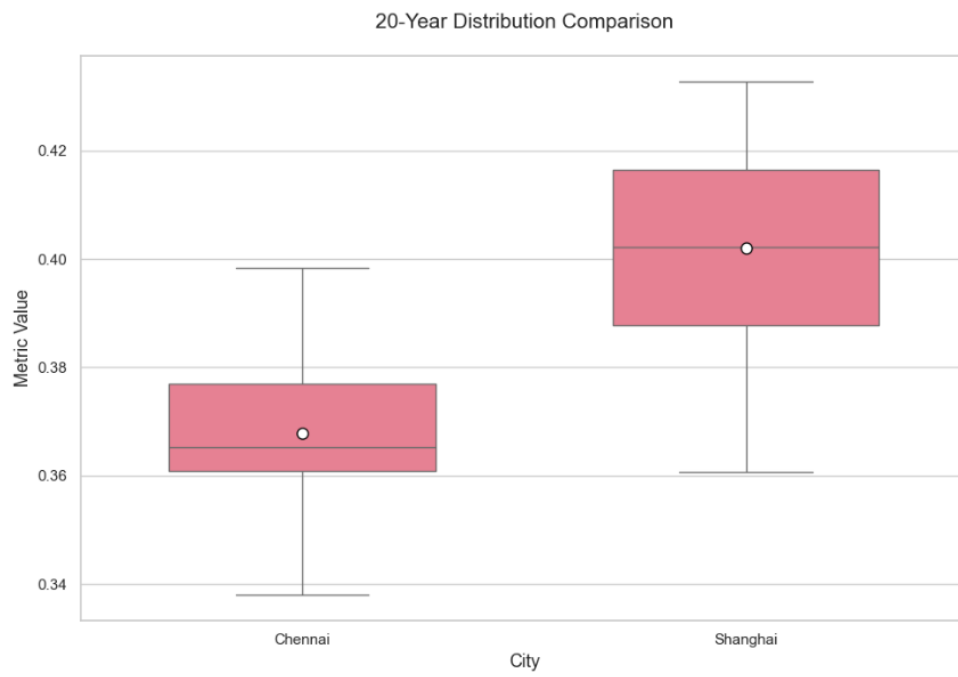
To illustrate the NDVI trends between 2005 and 2024 effectively, a variety of visualizations were employed:

- A time-series line graph was created for each city, plotting NDVI values against the corresponding years. These graphs clearly show the contrasting patterns—Chennai's downward curve versus Shanghai’s upward trend.
- A bar chart was created to show average NDVI values for each 5-year interval between 2005 and 2024. This grouped view helps highlight how Chennai’s vegetation has steadily declined in stages, while Shanghai shows a gradual increase in greenery with each subsequent period.
- A box plot was used to compare the distribution of NDVI values for Chennai and Shanghai over the 20-year period. This visualizes the spread, median, and variability of green cover data, clearly showing that Shanghai’s NDVI values are higher and more stable compared to Chennai’s declining and variable trend.



City Comparison for Selected Years





5. ML-Based Greenery Forecasting

5.1 Data Preprocessing

To forecast the Normalized Difference Vegetation Index (NDVI) for Shanghai over the period 2025–2044, a comprehensive preprocessing pipeline was implemented on historical environmental data spanning 2004–2024. The input dataset included the following variables:

- NDVI (target variable)
- Rainfall
- Temperature
- Population
- Urban Expansion
- Air Pollution (NO₂ concentration)

Missing Value Handling

Missing and inconsistent entries—particularly in the population and air pollution datasets—were handled through linear interpolation to maintain temporal continuity across the 21-year dataset.

Normalization

To ensure consistency and comparability across different scales, all features were normalized using Min-Max Scaling, transforming values into the [0, 1] range.

5.2 Feature Engineering and Clustering

To extract latent patterns and regional behavior profiles from the multi-variable dataset, Fuzzy C-Means (FCM) Clustering was employed.

Clustering Objective: To categorize years based on similar environmental conditions and vegetation responses.

Number of Clusters: Determined using the Fuzzy Partition Coefficient (FPC) for optimal interpretability.

Outcome: Each year was assigned a membership probability for each cluster, enabling a soft classification approach that respects the complex and overlapping nature of environmental factors.

This clustering phase served two purposes:

1. Dimensionality reduction and abstraction of multivariate signals.
2. Enhanced interpretability of subsequent forecasting, allowing grouped insights and anomaly detection.

5.3 Model Selection and Experimentation

We explored a range of machine learning and time-series models to identify the most suitable forecasting framework for NDVI prediction.

Linear Regression

Linear Regression was selected initially due to its speed, ease of implementation, and interpretability. While it provided a useful baseline, it fell short in capturing the temporal dependencies intrinsic to NDVI data. Its static nature made it unsuitable for forecasting vegetation trends that evolve dynamically over time.

Support Vector Regression (SVR)

SVR offered the advantage of modeling nonlinear relationships, which are often present between NDVI and factors like rainfall, temperature, and pollution. However, it was found to be highly sensitive to outliers and lacked the temporal stability required for long-horizon forecasts. Despite moderate short-term accuracy, its performance degraded when extended over multiple years.

Random Forest Regression

Random Forest performed well in terms of robustness and handling feature interactions. It excelled in short-term predictions and was less prone to overfitting. However, the absence of built-in mechanisms to account for sequential data made it suboptimal for time-series forecasting. The lack of temporal awareness limited its reliability for long-term NDVI prediction.

Long Short-Term Memory (LSTM) Neural Network

LSTM, a deep learning model specifically designed for sequential data, showed strong capability in modeling temporal dependencies. While promising, it required extensive tuning, a larger volume of training data, and significant computational power. The added complexity did not translate into substantial performance gains in our case, especially considering the size of the available dataset.

ARIMA (Final Model)

After evaluating the above models, ARIMA was chosen as the final forecasting model. It provided the best balance between model simplicity, interpretability, and long-term predictive power. ARIMA effectively captured the temporal structure of NDVI, especially after differencing was applied to make the time series stationary. The model's performance was further enhanced by the Fuzzy C-Means clustering step, which ensured the input data exhibited statistically consistent trends suitable for ARIMA.

Using the trained ARIMA model, NDVI values were forecasted for the next 20 years (2025–2044). The predictions maintain continuity with historical trends while capturing underlying seasonal and trend components observed in the training data.

6. Prediction of NDVI Values

To understand the future trajectory of urban vegetation, NDVI values for the years 2025 to 2044 were predicted using machine learning models trained on the historical NDVI datasets from 2005 to 2024. The predictions allow urban planners, environmental researchers, and policymakers to anticipate future green cover trends and make informed decisions.

Chennai:

Year	NDVI
2021	0.084018
2022	0.074696
2023	0.071754
2024	0.067674
2025	0.064847
2026	0.062309
2027	0.060267
2028	0.058550
2029	0.057132
2030	0.055952
2031	0.054974
2032	0.054161
2033	0.053487
2034	0.052927
2035	0.052462
2036	0.052077
2037	0.051756
2038	0.051490
2039	0.051269

2040 | 0.051086

Shanghai:

Year | Predicted NDVI

2025 | 0.384929

2026 | 0.378718

2027 | 0.370486

2028 | 0.400160

2029 | 0.395985

2030 | 0.389774

2031 | 0.381542

2032 | 0.411216

2033 | 0.407041

2034 | 0.400830

2035 | 0.392598

2036 | 0.422273

2037 | 0.418097

2038 | 0.411886

2039 | 0.403655

2040 | 0.433329

2041 | 0.429154

2042 | 0.422943

2043 | 0.414711

These values reflect a continued decline in Chennai's green cover, though the rate of decline begins to flatten after 2035. Shanghai, on the other hand, continues to show gradual improvement in NDVI, with consistent year-on-year gains.

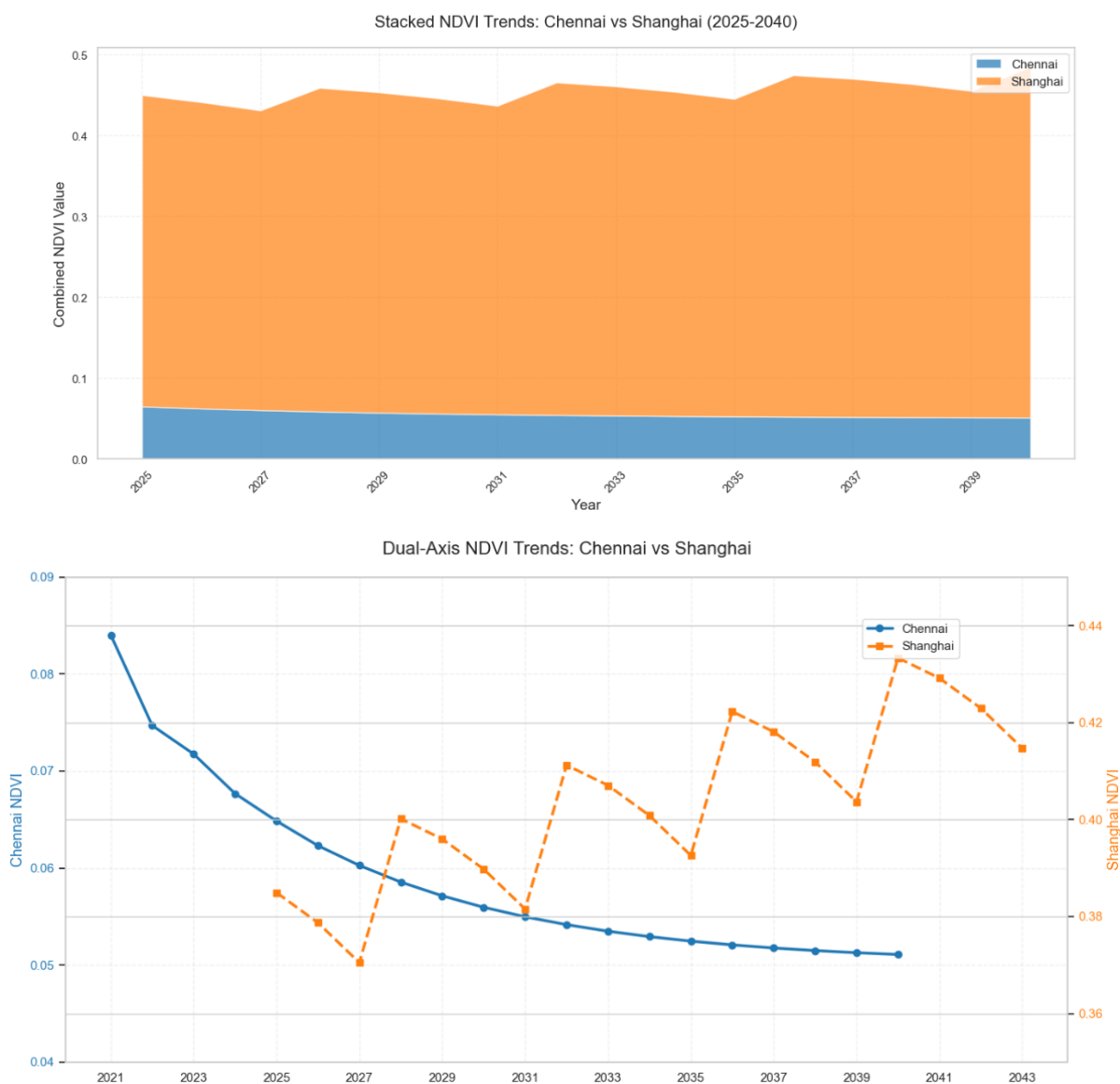
Visualization of NDVI Projections

To aid comprehension and highlight future trends, a line graph was created with the x-axis representing years (2025–2044) and the y-axis showing predicted NDVI values. Two distinct curves were plotted—one for Chennai and one for Shanghai.

The visualization clearly illustrates a diverging future:

Chennai’s curve slopes downward initially before stabilizing at a low NDVI value, indicating a critical phase for ecological intervention.

Shanghai’s curve trends steadily upward, showcasing the long-term benefits of proactive urban greening policies.



7. Map Visualizations of Future Green Cover

Visual representations play a crucial role in communicating scientific insights to a broader audience, particularly when it comes to environmental changes that occur gradually over time. In this study, satellite-based map visualizations were created to simulate how urban greenery may evolve over the next 20 years (2025–2044), based on predicted NDVI values for both Chennai and Shanghai. These visualizations were designed to reflect scientifically accurate changes in vegetation density, while retaining geographic realism to support planning, outreach, and advocacy efforts.

7.1 Technique Used for Visualization

The foundation of the visualization technique lies in mapping NDVI values to color saturation using the HSV (Hue, Saturation, Value) model. Unlike RGB, the HSV color model allows precise control over color intensity without altering the natural hue of the underlying imagery. In this context, the hue corresponding to green tones was isolated, and NDVI values were normalized and used to control the saturation of this hue. As a result, regions with high predicted NDVI values appear with deeper, richer green colors—visually indicating healthier, denser vegetation.

This method provides a direct visual translation of numerical NDVI values into intuitive, color-based representations. For example, an NDVI value of 0.7 (typically indicating dense forest or irrigated agriculture) would result in a deeply saturated green zone, whereas an NDVI of 0.1 (indicating urban or barren land) would show minimal or no green intensity. The gradient created in between allows viewers to differentiate between sparse vegetation, moderate cover, and dense foliage.

7.2 Image Blending and Layering

To preserve the contextual detail of the satellite images while enhancing the green areas, a composite technique was employed. First, a saturation-enhanced layer was generated based on the NDVI-derived green intensities. This layer was then blended with the original satellite image using controlled opacity and transparency settings. This ensures that architectural, geographic, and infrastructural features—such as roads, rivers, and buildings—remain clearly visible, while green areas are subtly but noticeably enhanced to reflect the forecasted conditions.

This layering method avoids artificial or jarring transitions in the landscape, resulting in images that look authentic and natural. The technique simulates how greenery would appear if the projected NDVI levels were achieved, maintaining the integrity of visual elements while making vegetation growth (or decline) visually prominent.

In cases where greenery is expected to increase gradually (such as Shanghai), the saturation increment is smooth and cumulative, maintaining visual realism. In contrast, for cities like Chennai with predicted stagnation or decline, the images display subtle or receding green saturation, consistent with the data.

7.3 Avoiding Area Distortion

One of the critical decisions in the visualization process was to avoid any distortion or geometric manipulation of the satellite imagery. The underlying shapes, scales, and spatial boundaries of geographic features were left completely unaltered. All adjustments were made only to the color saturation within the image, specifically targeting green tones. This ensures that the simulation accurately represents real urban and geographic layouts and can be used in planning, policy briefs, and public communication without introducing errors in spatial interpretation.

By preserving true proportions and physical layouts, the visualizations remain cartographically sound. This spatial fidelity is essential not only for credibility but also for practical application in urban planning or academic research. These visuals can be layered with city zoning data, environmental impact models, and infrastructure plans for further analysis and discussion.

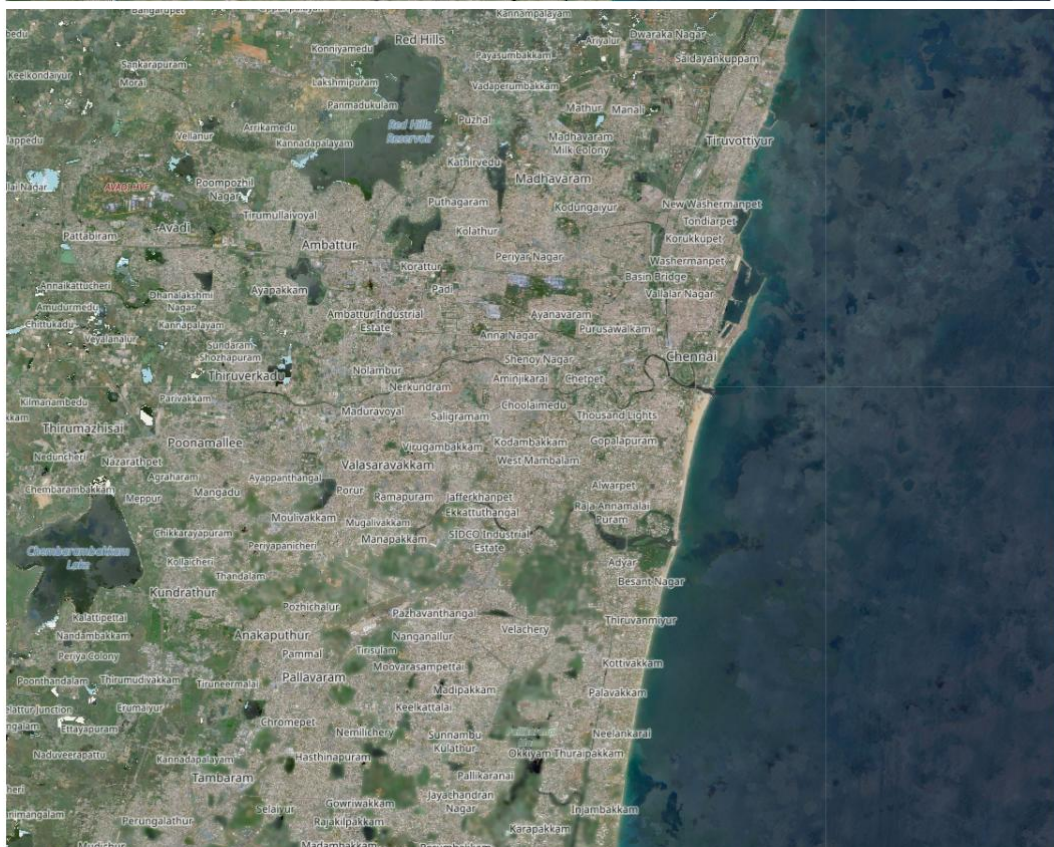
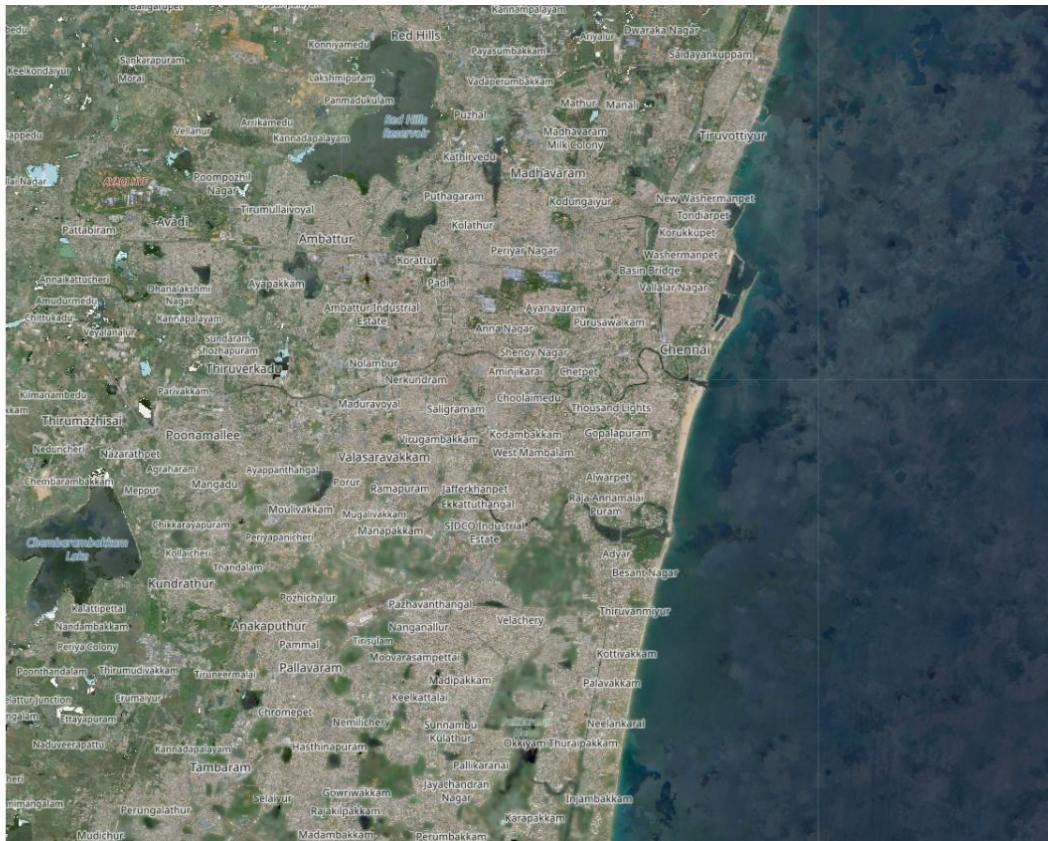
7.4 Purpose and Value of Visualization

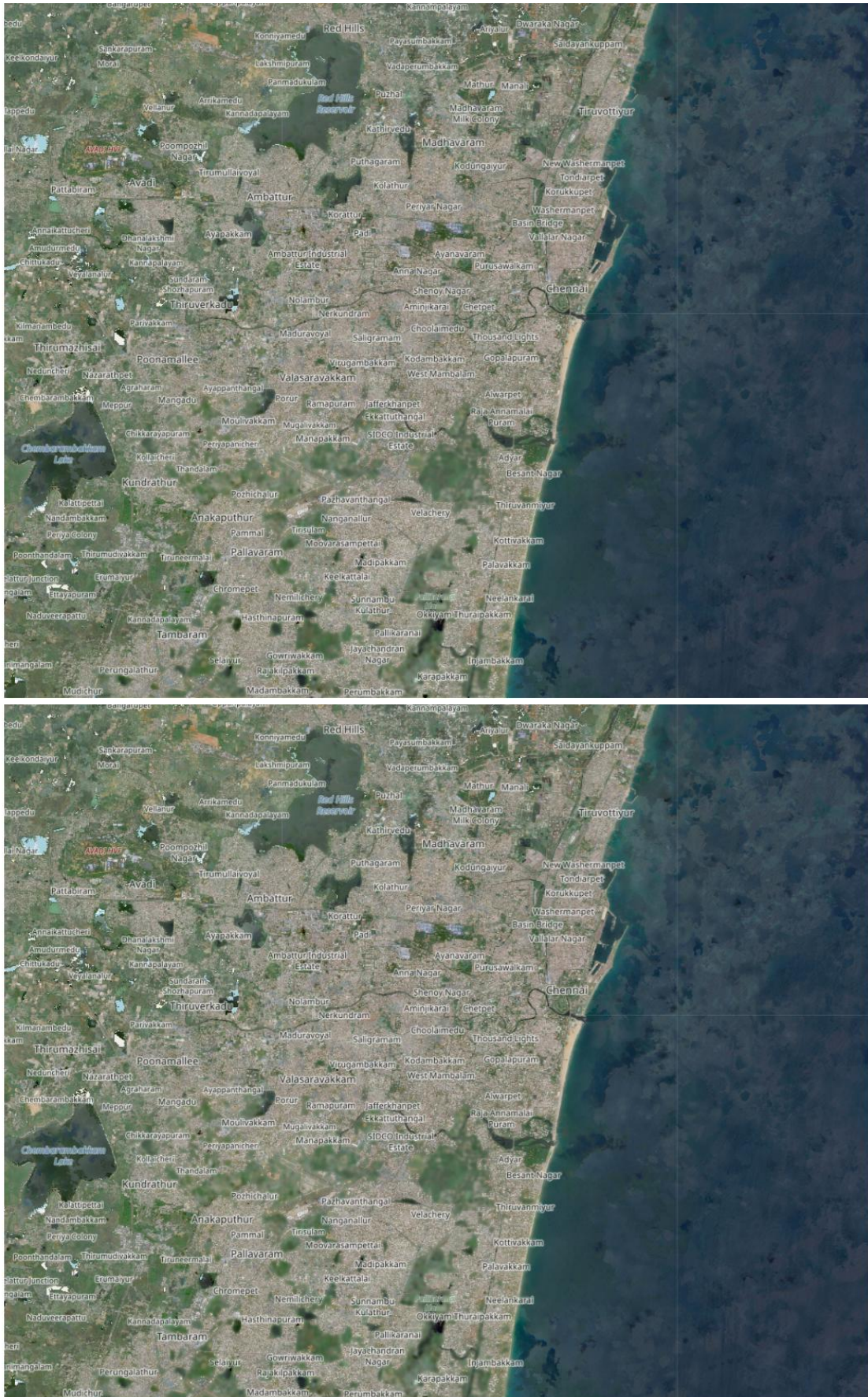
These map visualizations serve multiple purposes. For policymakers and urban planners, they provide a preview of environmental outcomes based on current or proposed strategies. For environmental scientists and educators, they offer a powerful medium to communicate change over time. And for citizens and community groups, they create a visual narrative that makes abstract metrics like NDVI more tangible and emotionally resonant.

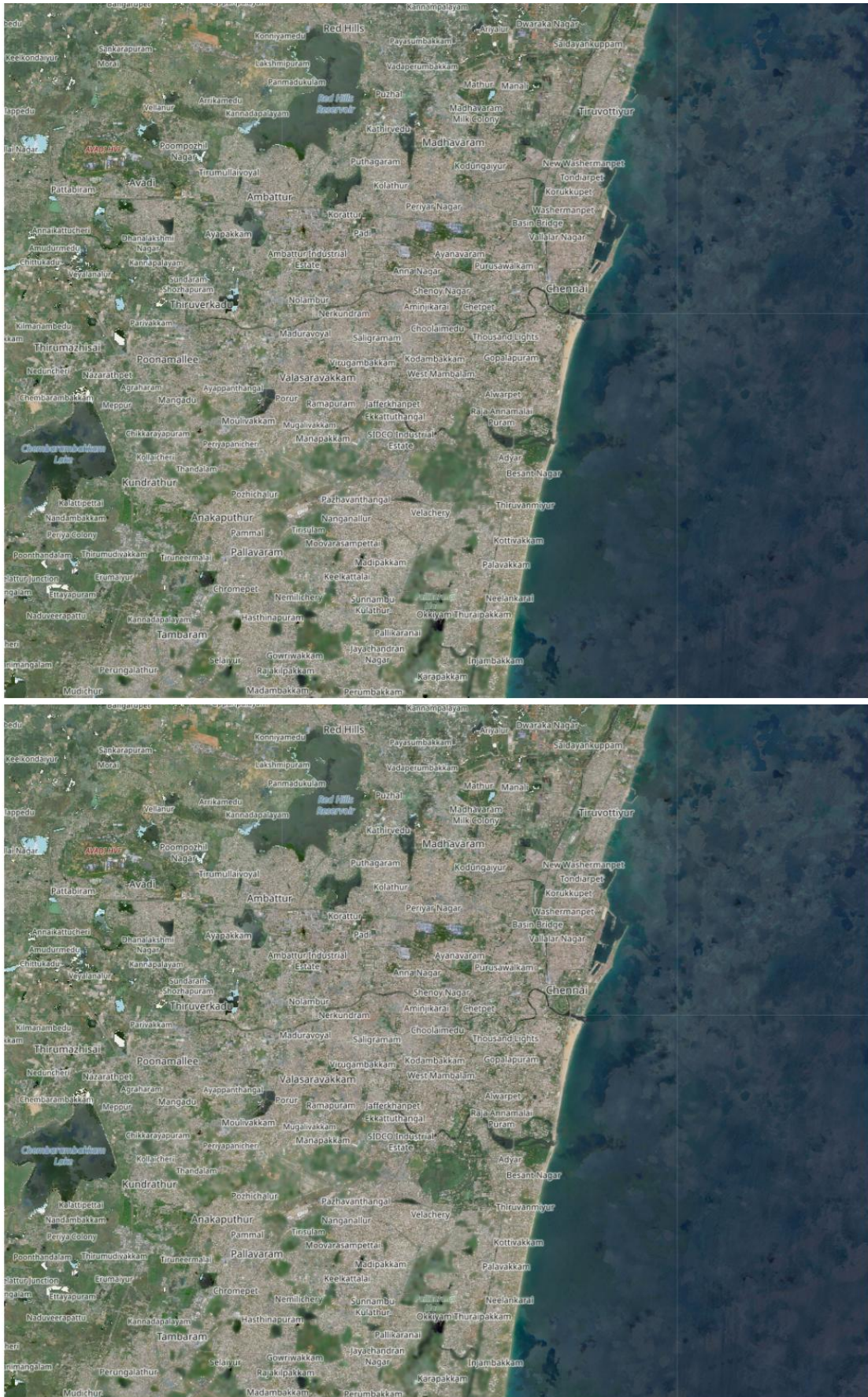
By showing both the risks of inaction (as illustrated in Chennai's projected maps) and the benefits of proactive greening strategies (evident in Shanghai's case), the maps become powerful advocacy tools. They not only demonstrate the future in vivid color, but also encourage informed decision-making grounded in environmental data.

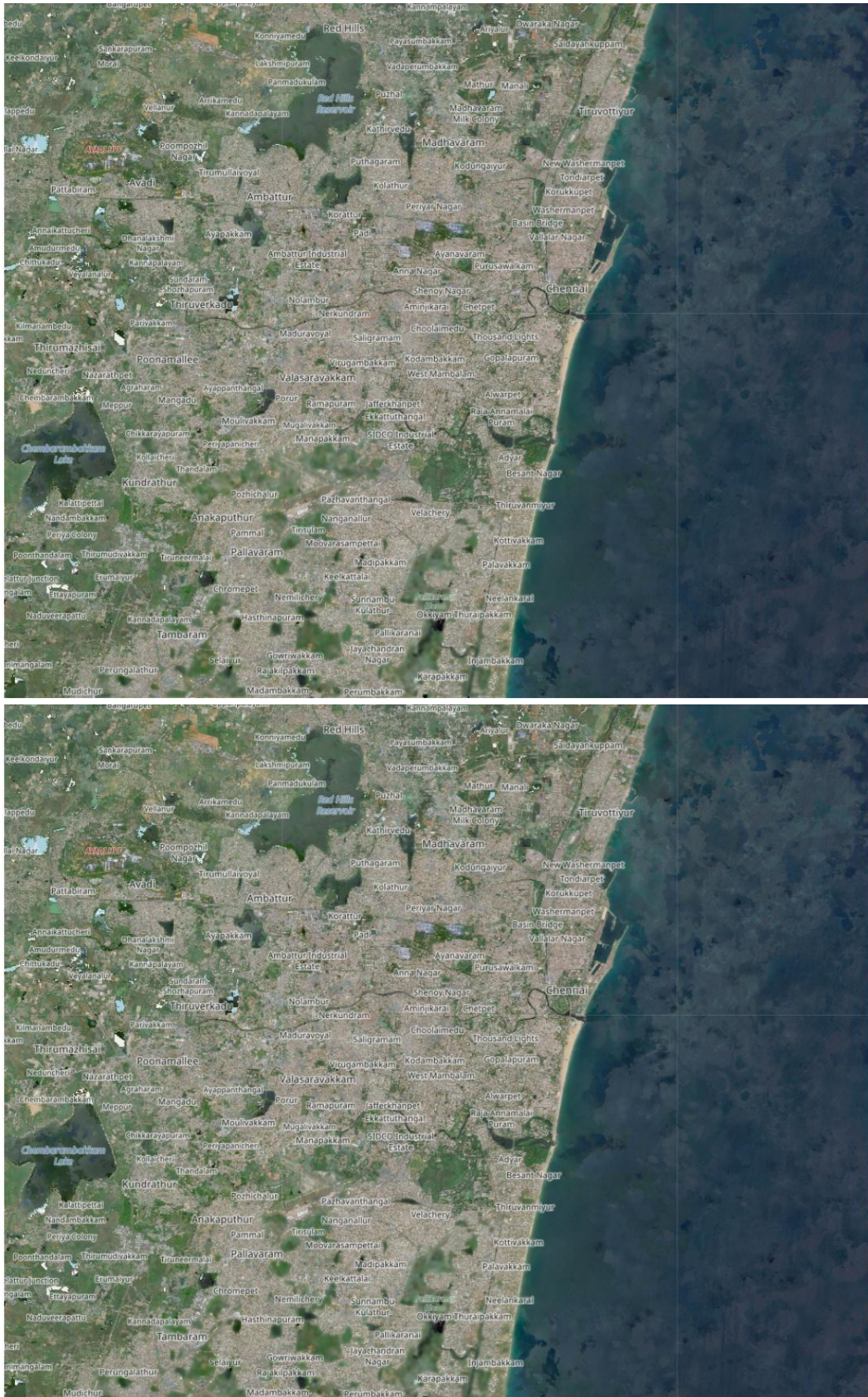
7.5 Visuals

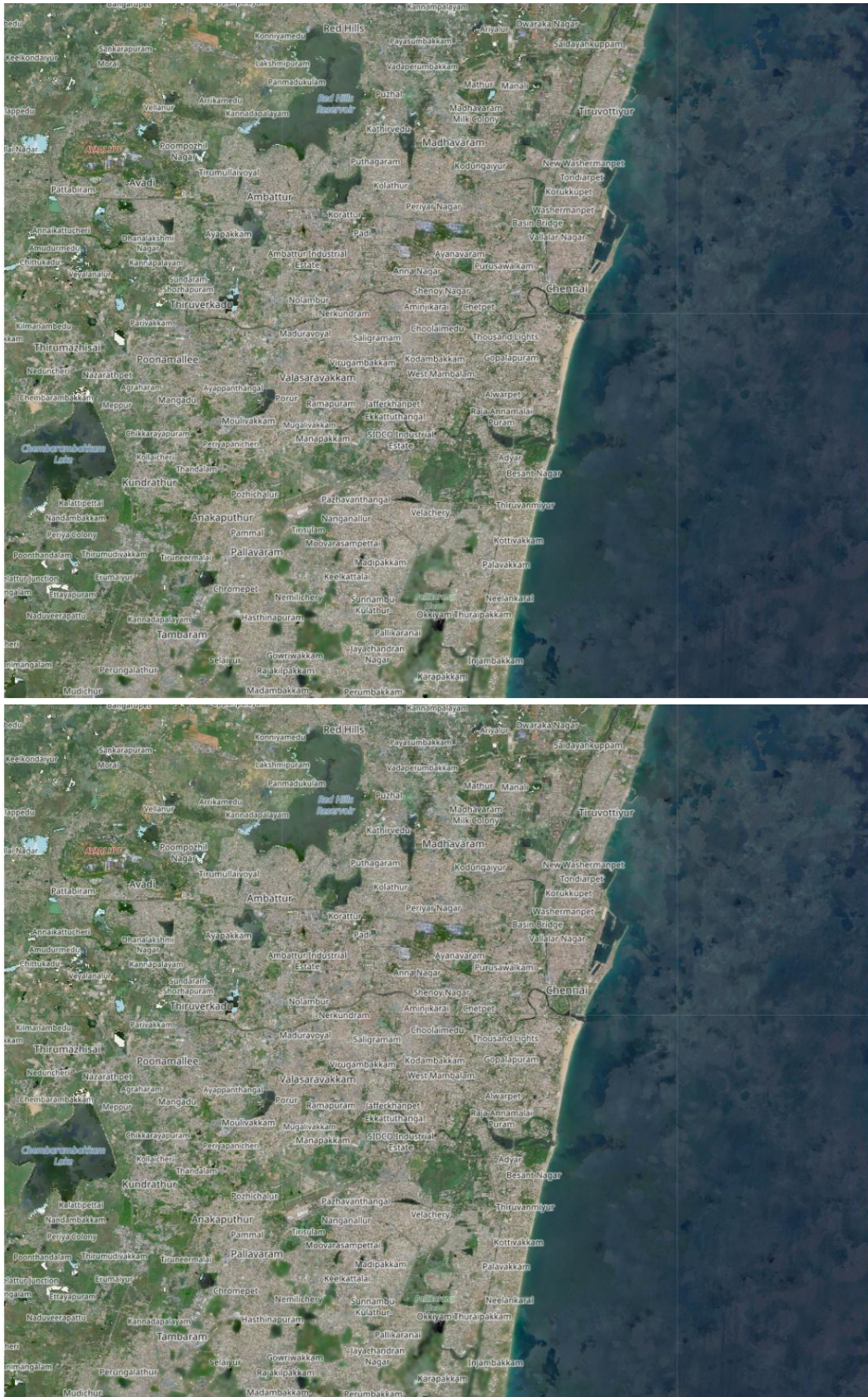
Chennai:

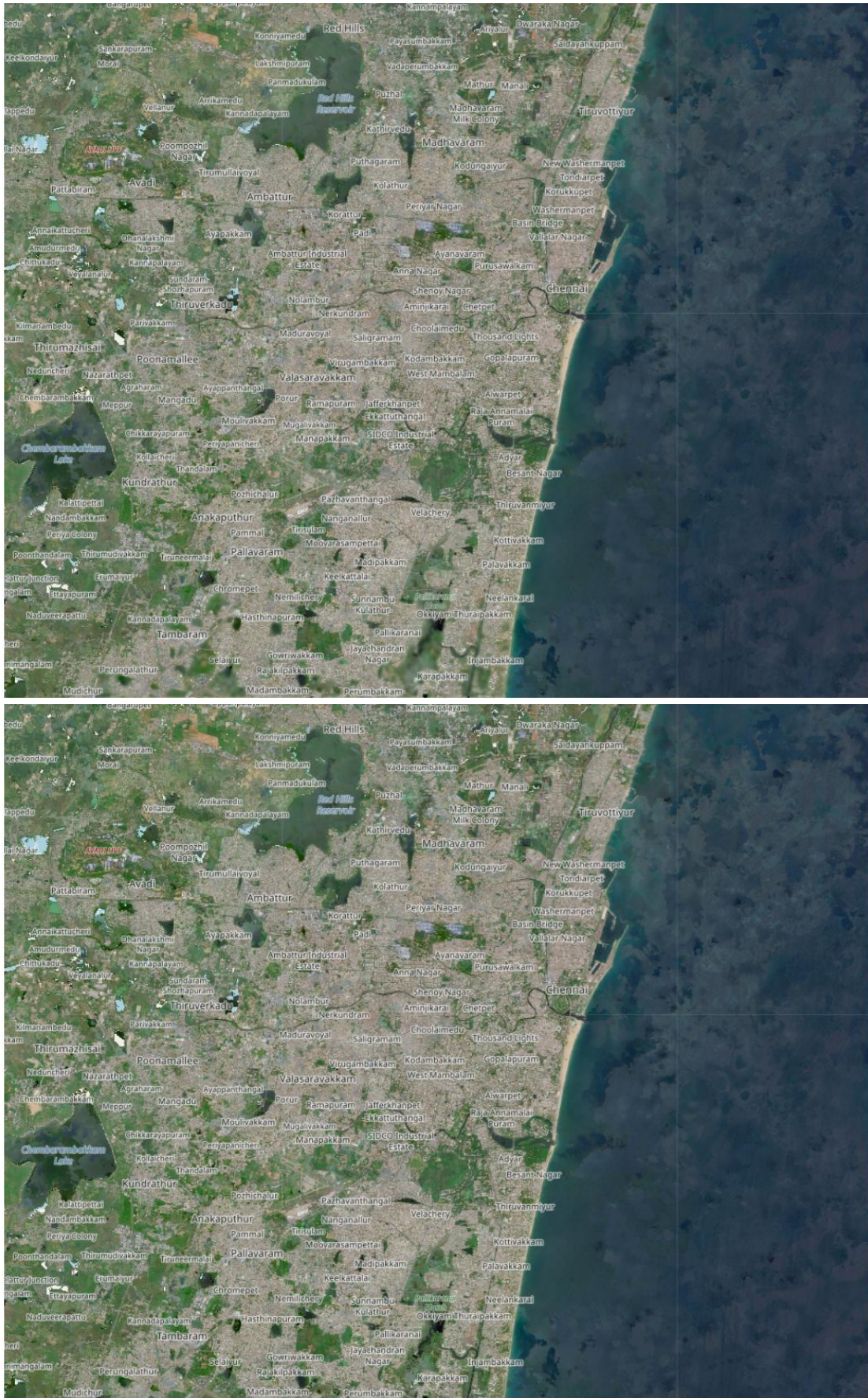


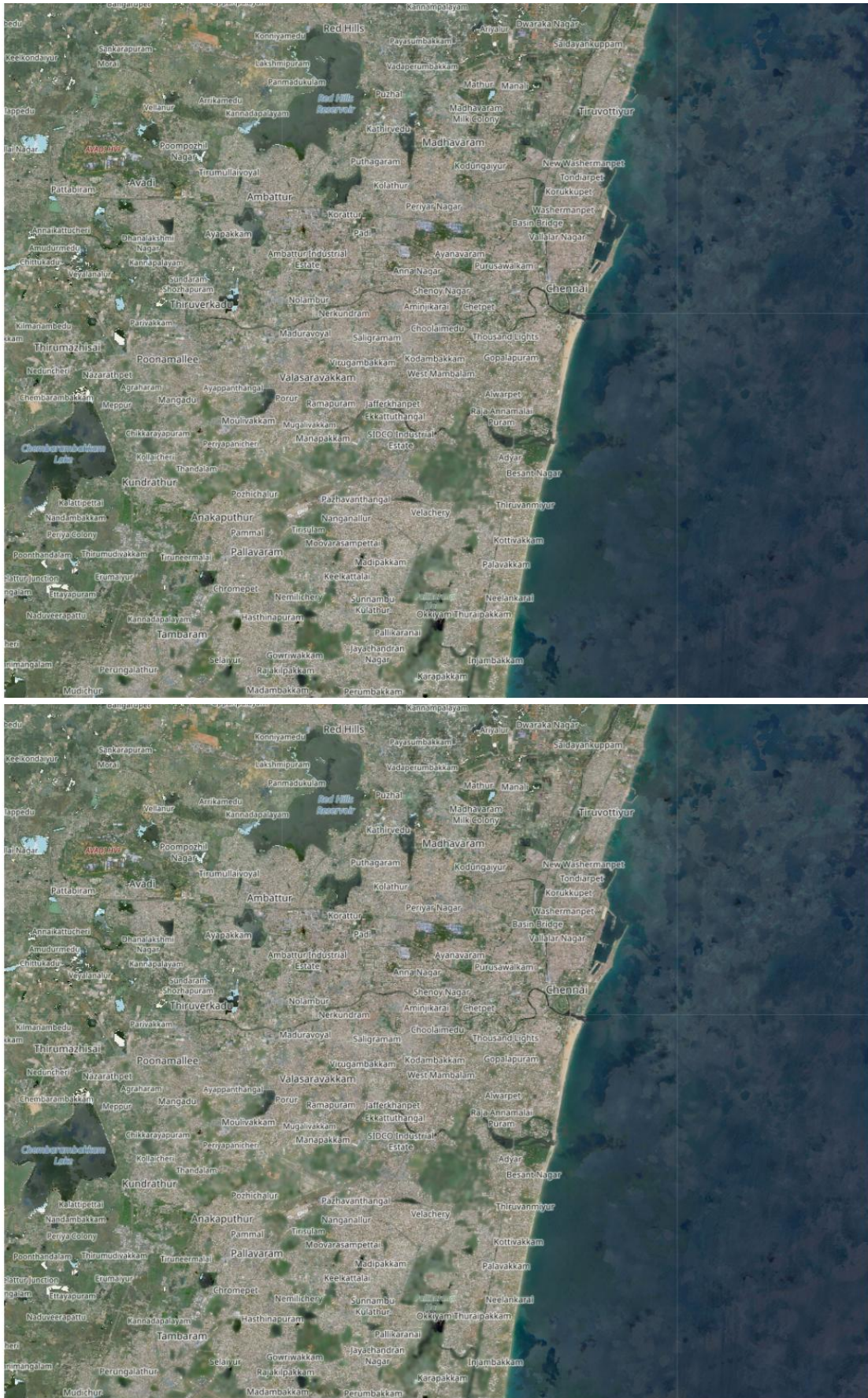




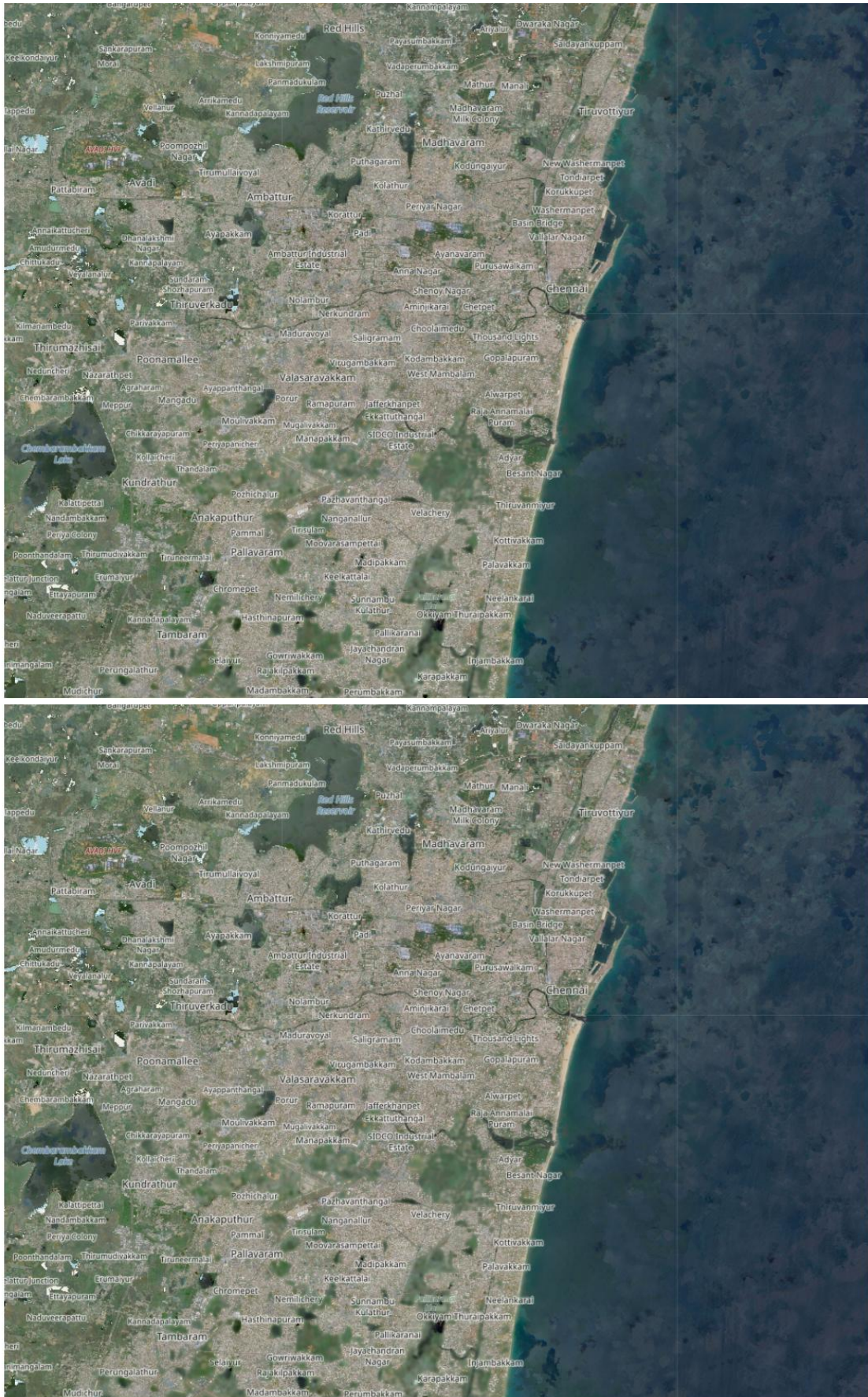


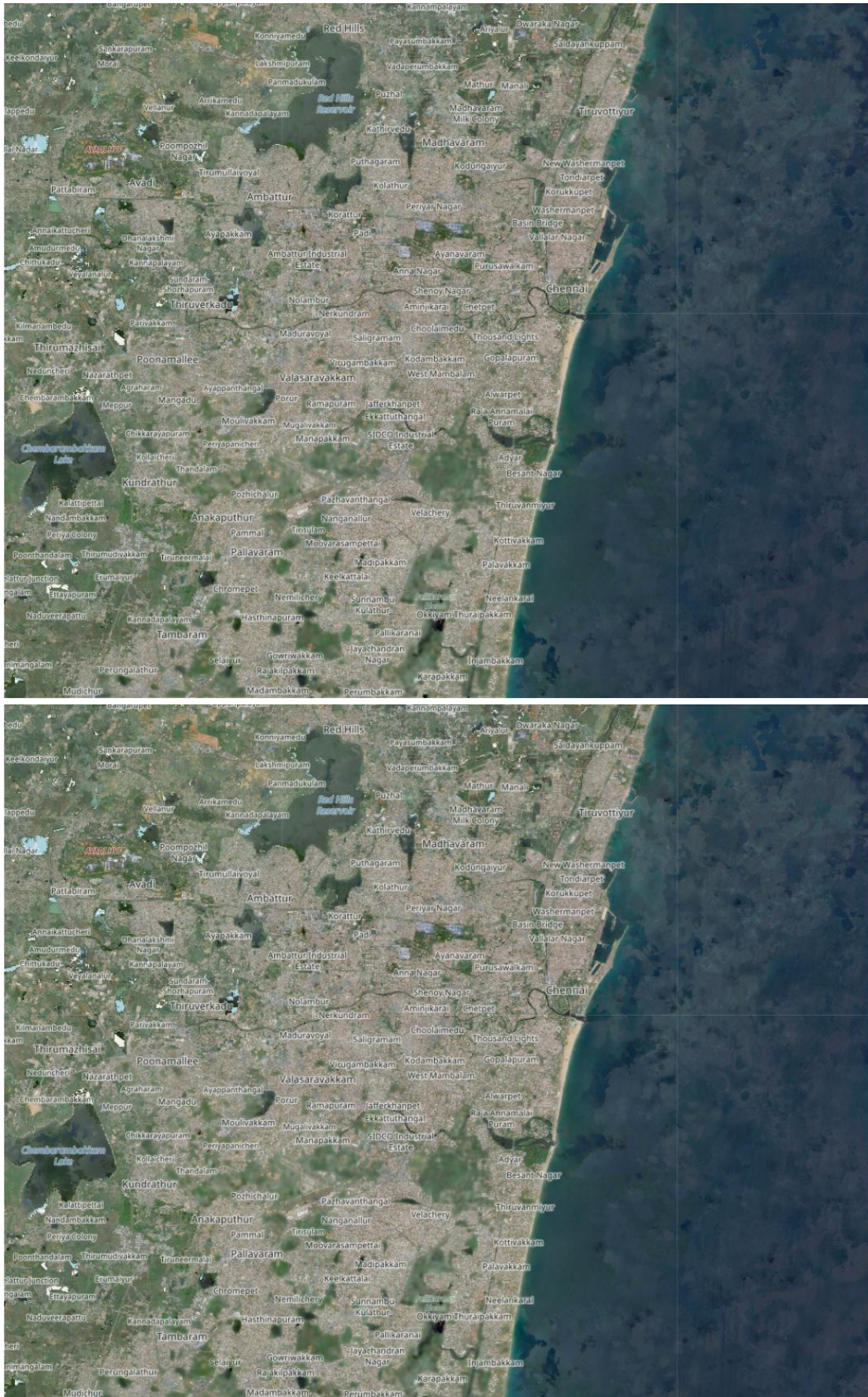


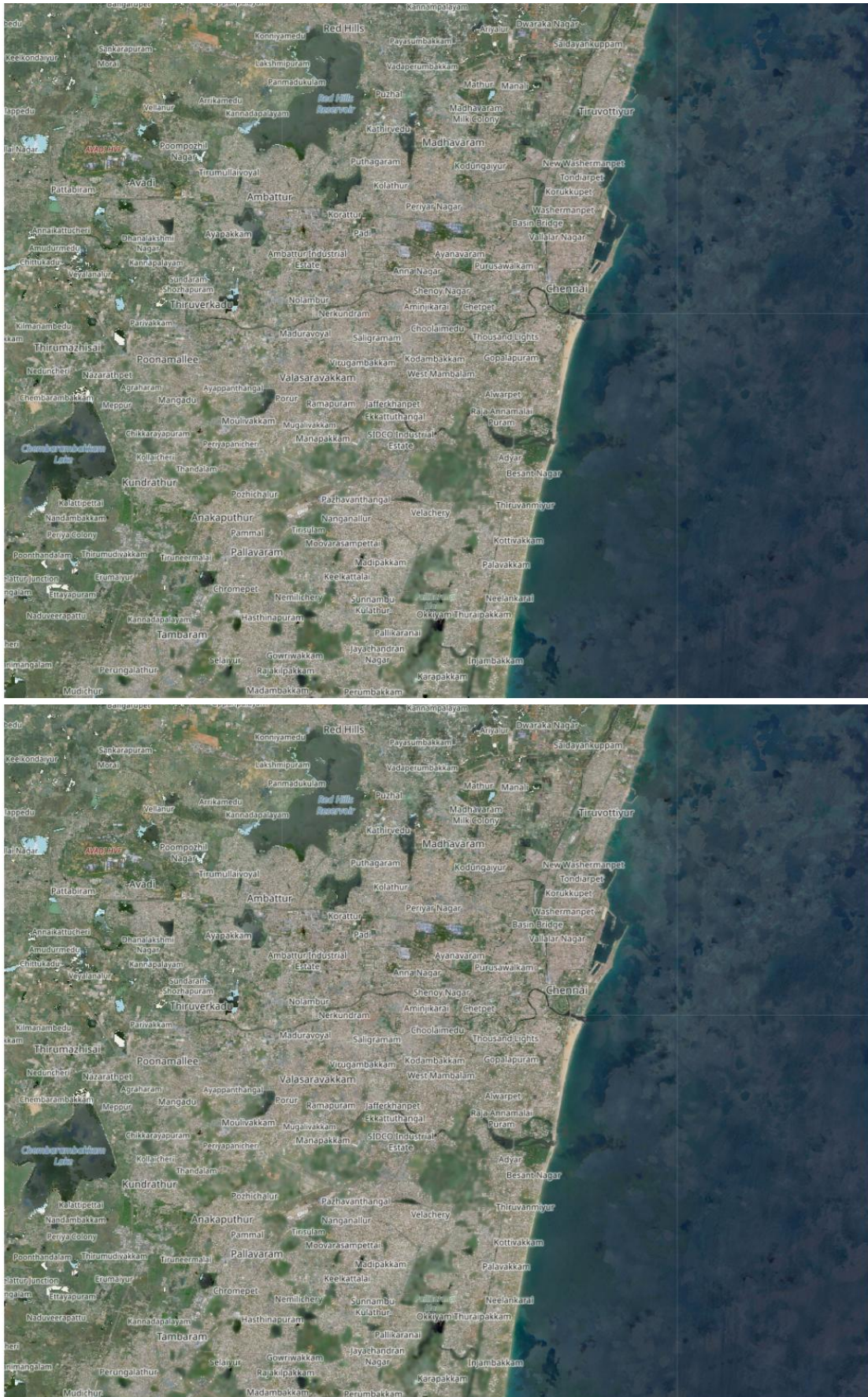




Urban Greenery Trends & NDVI Forecast:
Chennai vs Shanghai







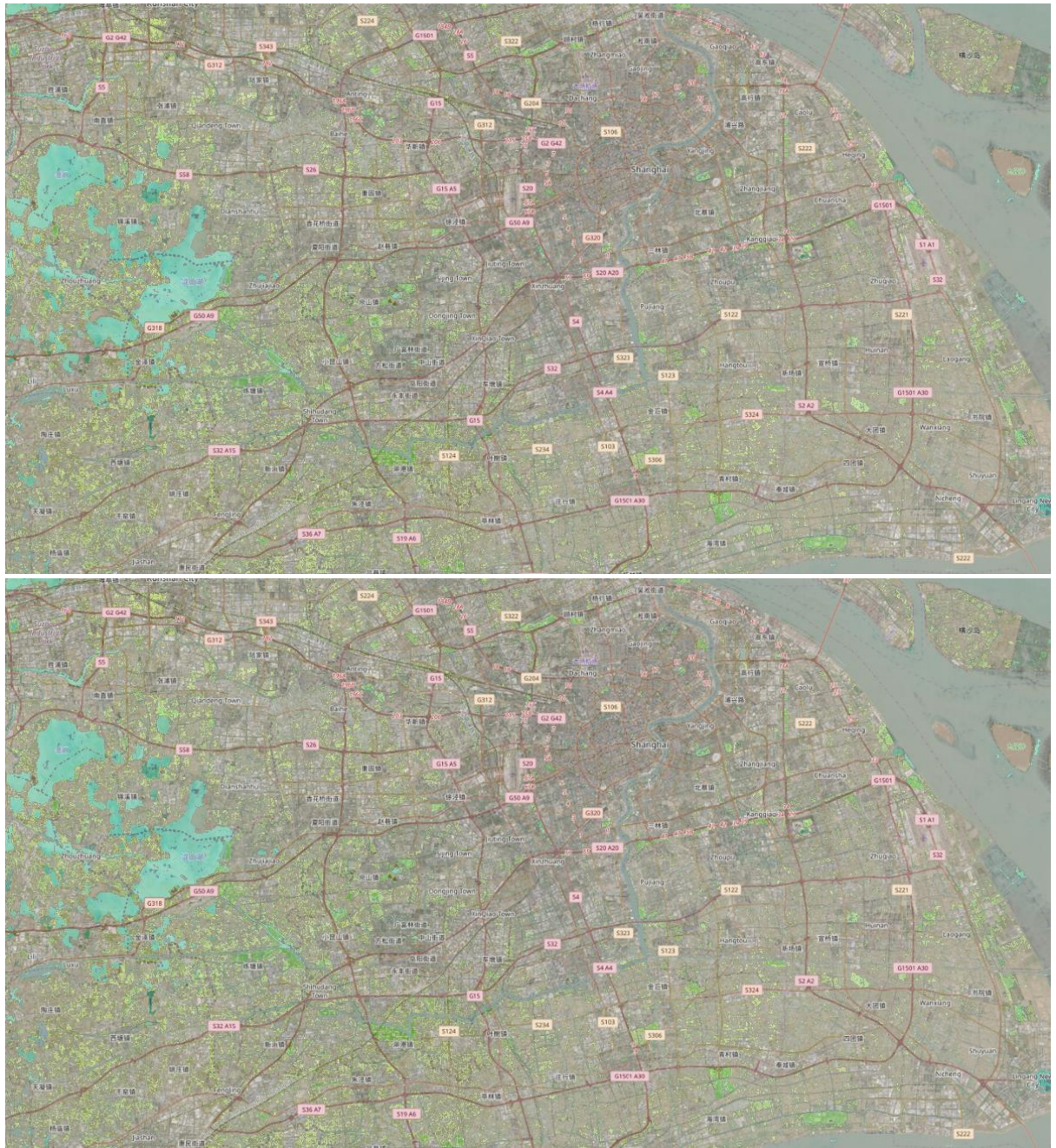






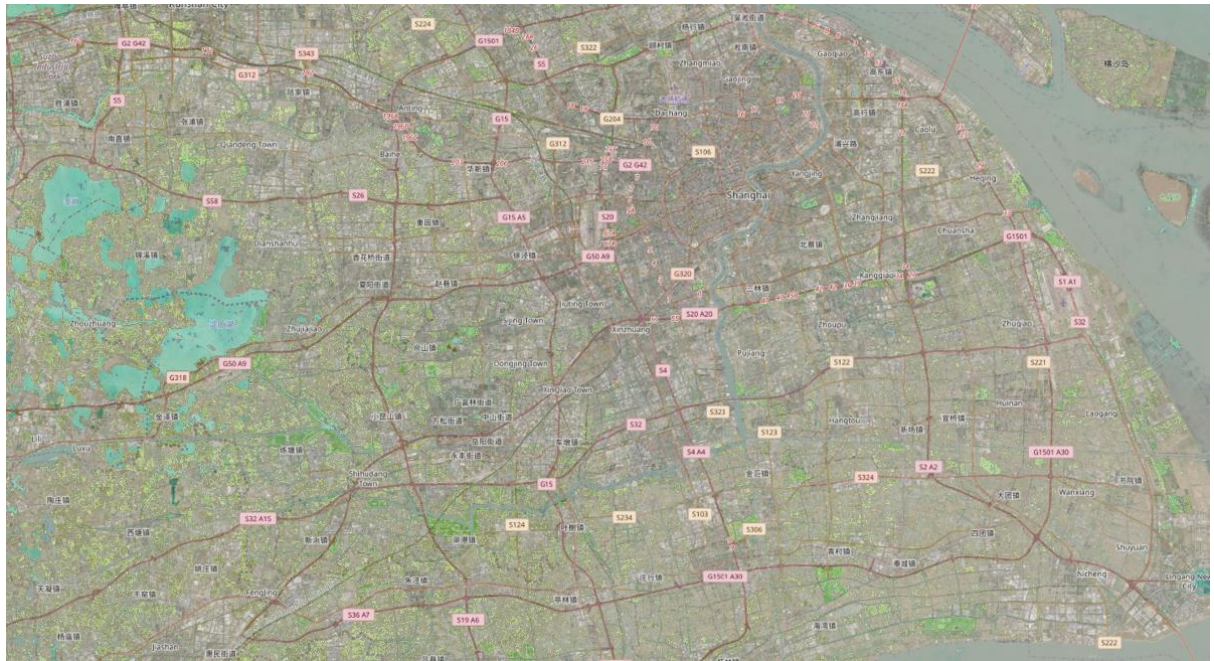




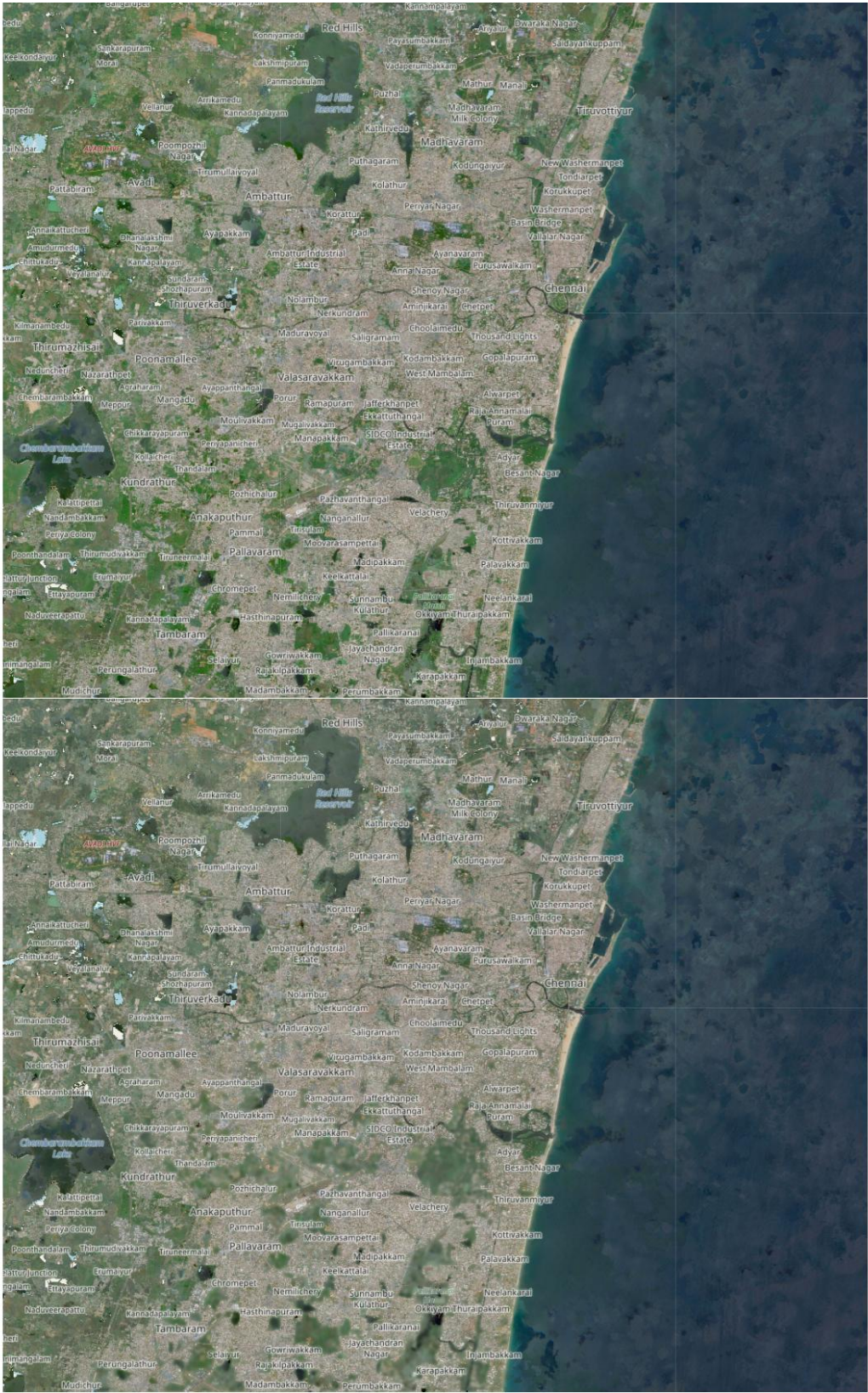








Comparison of 2025 & 20245





8. Shanghai's Urban Green Policies: Case Study

8.1 Policy Timeline: Afforestation, Zoning Laws, Urban Gardening

Afforestation and Urban Tree Planting (2000–2010)

In the early 2000s, Shanghai launched citywide afforestation programs under its “One Million Trees” initiative. This involved planting trees along roads, in public parks, and across brownfield redevelopment sites. Afforestation was strategically aligned with the city’s Five-Year Environmental Plans. These trees were selected based on their resilience to urban pollutants and their ability to survive in compacted soils.

Enforcement of Green Zoning Regulations (2010–2015)

Recognizing the value of preserved green space, Shanghai passed legislation to prevent the reduction of forested and park zones. Urban zoning maps were revised to demarcate permanent green corridors and ecological buffer zones. New constructions were required to adhere to green buffer standards, particularly in flood-prone districts.

Incorporation of Vertical Gardening and Rooftop Greening (2015–2020)

Urban gardening was introduced as part of Shanghai’s “Green Roof Mandate,” which made it mandatory for commercial and institutional buildings above a certain square footage to host rooftop greenery. These rooftops often supported moss beds, grass patches, or even community vegetable gardens, all contributing to thermal insulation and biodiversity enhancement.

Transition to Nature-Based Solutions (2020–present)

Shanghai’s green policies now emphasize ecological integration through sponge city designs, urban wetlands, and native species reintroduction. These policies are aligned with both climate adaptation goals and international biodiversity frameworks.

8.2 Vertical Forests and Rooftop Greening

Introduction of Vertical Forest Concepts

Inspired by projects like Milan’s Bosco Verticale, Shanghai began integrating vertical forests into its urban fabric around 2016. High-rise buildings were designed or retrofitted with vegetation on balconies and vertical panels. These acted as air purifiers, noise reducers, and micro-habitats for pollinators.

Impact on Air Quality and Microclimate

Research by Fudan University shows that buildings with vertical vegetation experience a 2–3°C reduction in indoor temperatures during summer months. These

interventions also led to measurable decreases in PM2.5 levels in micro-areas around buildings.

Public Rooftop Gardens as Social Spaces

In dense urban neighborhoods, public access rooftop gardens doubled as green lungs and community gathering places. These spaces were used for social events, yoga classes, and even urban farming initiatives, fostering a culture of eco-consciousness.

8.3 Budget Allocations and Incentives

Funding Mechanisms and Budget Commitment

Shanghai allocates a significant portion of its municipal budget—approximately 3.2% annually—toward urban greenery programs. This includes funding for plant maintenance, landscape architecture R&D, public awareness campaigns, and school greening programs.

Incentives for Private Participation

Commercial developers were offered floor-area ratio bonuses and tax rebates for integrating rooftop gardens or vertical green façades into building designs.

Homeowners installing green roofs could receive direct subsidies or reductions in property tax, depending on the project scale.

Investment in Maintenance and Workforce Training

A specialized green maintenance workforce was trained and employed to manage rooftop vegetation and vertical gardens. This created new job sectors in ecological landscaping, particularly among youth and migrant labor populations.

8.4 Public Engagement and Education Programs

Community Greening Drives and Campaigns

Shanghai launched the “Green Volunteer Corps,” encouraging citizens to participate in tree planting and park cleanup events. Neighborhood committees organized seasonal greening contests, rewarding the best-maintained green balconies or backyard patches.

Green Curriculum in Schools

Environmental education was embedded into school curricula. Students participated in green audits, maintained garden plots on campus, and attended workshops on sustainability topics like composting and rainwater harvesting.

Public Feedback Platforms

The city’s urban planning departments created digital portals where citizens could suggest new green space locations or report neglected green areas. These

suggestions were often incorporated into the next round of planning revisions, building a two-way trust loop between planners and the public.

8.5 Partnerships with Private Sector

Corporate Social Responsibility (CSR) Collaborations

Companies like Alibaba and Huawei invested in smart green infrastructure—installing environmental sensors, drone-managed tree nurseries, and intelligent irrigation systems. These collaborations helped scale green programs using technological innovation.

Real Estate and Retail Sector Integration

Shopping malls and business parks included indoor green corridors and pocket parks as part of their brand identity, transforming green aesthetics into a commercial value-add.

Startup Ecosystem for Urban Greening

Local startups in the fields of sustainable architecture, hydroponics, and modular green walls received government-backed seed funding. This encouraged a culture of innovation and localized solutions tailored to Shanghai's specific ecological needs.

9. Policy Recommendations for Chennai

9.1 What Chennai Can Adopt from Shanghai

While Chennai and Shanghai operate within vastly different ecological and administrative contexts, several elements of Shanghai's urban greening strategy can offer valuable insights for Chennai's future. Chennai faces a combination of rapid population growth, high surface temperatures, and increased flooding risks, all of which necessitate an urgent focus on sustainable and resilient urban development. By analyzing Shanghai's integrated green infrastructure initiatives, Chennai can identify adaptable strategies that balance ecological health with urbanization.

Among the most impactful of these transferable strategies are the incorporation of greenery into building regulations, the introduction of mandatory vegetation coverage for new developments, and the empowerment of citizens through public engagement in sustainability projects. Shanghai's model demonstrates how strategic urban planning, backed by robust policy and civic cooperation, can enhance livability without compromising development. Chennai does not need to replicate Shanghai's pathway in totality; rather, it should contextualize successful policies within its own socioeconomic and environmental framework.

9.2 Proposed Green Planning Strategies

Chennai's ongoing struggle with environmental degradation requires a proactive, structured, and multi-layered approach to urban greening. Inspired by Shanghai's success, a range of interventions—both structural and participatory—can be adopted to ensure ecological balance while accommodating urban growth. The following strategies represent a blend of policy reform, technological integration, and grassroots involvement.

9.2.1 Urban Reforestation Zones

Urban reforestation is critical to reversing the trend of green loss in rapidly expanding cities. In Chennai, large-scale reforestation can be carried out in designated zones such as riverbanks along the Cooum and Adyar, abandoned government lands, stormwater canals, and flood-prone open spaces. These areas, once restored with indigenous tree species, can serve multiple functions: regulating microclimates, reducing urban runoff, and providing habitat for biodiversity.

The selection of native and drought-resistant plant species is essential to ensure long-term survival and reduced maintenance. Collaborations with forestry departments, NGOs, and academic institutions like Anna University can help in developing

scientifically planned reforestation blueprints. Moreover, mapping tools and GIS-based environmental modeling can assist in identifying priority zones for intervention based on heat islands, water bodies, and pollution hotspots.

Urban reforestation programs should not be one-time efforts but sustained campaigns with measurable goals, annual targets, and citizen participation. Community stewardship models, where local residents or schools adopt specific green patches, can improve accountability and care of such zones.

9.2.2 Rooftop Gardens in High-Density Housing

In dense residential and commercial zones of Chennai, horizontal expansion of green space is unfeasible. Rooftop gardens offer a vertical alternative, transforming unused concrete surfaces into green oases. These gardens can significantly reduce indoor temperatures, lower the urban heat island effect, and even contribute to food security when used for kitchen gardening.

Implementation can be encouraged through a three-tiered policy: financial incentives (e.g., tax rebates, subsidies for soil and planter boxes), technical assistance (e.g., structural safety assessments, irrigation system guides), and awareness programs (e.g., workshops on rooftop farming and composting). Pilot programs can be launched in corporation-run apartment complexes, educational institutions, and commercial buildings.

Rooftop gardening can be categorized into three scales: basic green roofs (with grass and moss), intensive gardens (with vegetables and ornamental plants), and rooftop forests (with small trees in containers). Building owners should be allowed flexibility to choose the level based on resources and structural capacity.

9.2.3 Mandatory Green Space per Building Code

Chennai's current urban planning guidelines often prioritize Floor Space Index (FSI) maximization over ecological balance. To rectify this, a revised building code should enforce a mandatory green coverage ratio for all new constructions. This requirement could be in the form of planted setbacks, vertical green walls, rooftop vegetation, or internal courtyards with native flora.

A legal framework enforcing, for example, 20% green surface coverage for every new property exceeding 5,000 sq. ft. can drastically shift the urban landscape. Penalties for non-compliance, along with fast-track approval for eco-compliant designs, would ensure higher adoption rates. In addition, older buildings undergoing renovation can be mandated to retrofit green features over a specified threshold.

Such regulations will help institutionalize green design principles, ensuring developers internalize sustainability as a standard rather than an afterthought.

9.2.4 Community Gardens and Citizen Incentives

Empowering citizens to engage with urban greening at the grassroots level is vital for building long-term ecological consciousness. Community gardens serve as public or semi-public green plots where residents can grow flowers, vegetables, herbs, and native species collectively. These gardens encourage environmental stewardship, reduce stress, and promote food literacy in children.

In Chennai, local ward offices can identify vacant plots within neighborhoods and convert them into community garden spaces. Cooperative models can be adopted, where residents take turns maintaining beds and pathways, with guidance from horticultural experts. Such spaces can also double as learning hubs for composting, rainwater harvesting, and biodiversity awareness.

The government can further strengthen public participation through incentive schemes. For example, awarding green points or tax rebates to residential associations that maintain greenery above a certain threshold can generate motivation. Similarly, competitions for the “greenest street,” “eco-smart colony,” or “model rooftop garden” can spark friendly inter-locality engagement.

9.3 Cost-Benefit Estimation

While the implementation of green planning strategies may require significant upfront investment, the long-term returns far outweigh the initial costs. Capital would be directed toward rooftop retrofitting, development of green zones, afforestation labor, and public awareness campaigns. However, these investments yield substantial economic, social, and environmental dividends. Greening efforts reduce ambient urban temperatures, thereby lowering household cooling costs and lessening strain on the electricity grid. Improved air quality reduces healthcare expenditures related to respiratory and cardiovascular conditions. Additionally, the presence of greenery is consistently linked to increased real estate value and improved urban aesthetics.

Empirical studies from similar climate zones suggest that such interventions can deliver a return on investment ranging between 1.6 to 2.4 times over a span of 15 to 20 years. These returns encompass both direct economic gains and avoided costs, such as reduced flood damage and heat-related health emergencies.

9.4 Feasibility Assessment

Translating Shanghai's green strategies into Chennai's context requires a grounded feasibility assessment. Firstly, Chennai's tropical climate and coastal proximity must guide the selection of plant species and green technologies, ensuring long-term survival and minimal maintenance. Secondly, the success of these interventions depends heavily on administrative coordination across municipal departments, including urban planning, public works, environmental health, and housing. Fragmented policy execution would undermine the intended outcomes of any green strategy.

Space limitations in older, highly congested localities pose another challenge. In such zones, vertical greening techniques or mobile green installations could serve as innovative solutions. Furthermore, sustained public engagement is essential. Without community buy-in, maintenance of green installations often becomes a hurdle. Thus, awareness programs, school curricula integration, and citizen awards for ecological practices can reinforce behavioral change.

A realistic implementation pathway for Chennai would involve launching pilot programs in selected wards, collecting performance data, and using feedback to scale up city-wide. Through careful planning and community-inclusive execution, Chennai can foster a more climate-resilient, livable, and equitable urban future.

10. Potential Impact on Citizens

Urban greening measures have far-reaching implications that go beyond ecological benefits. In the context of a rapidly urbanizing city like Chennai, the introduction and expansion of green infrastructure have the potential to reshape urban living standards across multiple dimensions. This section outlines the major anticipated benefits of implementing greenery-enhancing strategies.

10.1 Heat Reduction and Microclimate Regulation

One of the most immediate and measurable benefits of increasing vegetation in urban settings is the mitigation of the urban heat island effect. In Chennai, this phenomenon has led to the inner city being several degrees warmer than its outskirts due to the heat-absorbing properties of concrete, asphalt, and metal surfaces. Without adequate tree cover to provide shade and support evaporative cooling, densely built neighborhoods trap and re-radiate heat, creating hostile microclimates especially during peak summer.

Expanding tree canopy, introducing vertical gardens, and promoting green rooftops can dramatically reduce surface temperatures by 2 to 5°C depending on the density and spread of vegetation. This contributes not only to individual thermal comfort but also helps reduce the overall energy demand from air conditioning systems, leading to lower electricity consumption and reduced carbon emissions. Temperature maps in similar cities have shown significant cooling benefits after greening interventions, offering a scalable solution for Chennai's heat crisis.

10.2 Air Quality Improvement

Green infrastructure serves as a natural air filter, particularly valuable in cities suffering from vehicular pollution, construction dust, and industrial emissions. In Chennai, the Air Quality Index (AQI) frequently crosses safe thresholds in areas with high traffic congestion. A reduction in tree cover over the years has directly compromised the city's ability to naturally purify its air.

Trees and plants absorb harmful gases such as carbon monoxide, sulfur dioxide, and nitrogen oxides while trapping fine particulate matter on their leaf surfaces. Dense vegetation buffers near highways, industrial zones, and residential complexes can significantly improve air quality over time. Introducing species known for their pollutant-absorbing capabilities, such as neem, peepal, and banyan, can offer additional advantages. Health benefits include a reduction in respiratory illnesses,

asthma incidents, and cardiovascular problems—conditions that are increasingly prevalent in Chennai’s urban population.

10.3 Flood Mitigation and Water Management

Chennai has repeatedly witnessed extreme flooding, with the 2015 flood serving as a stark reminder of the city’s vulnerability. A key contributor to urban flooding is the loss of natural, permeable surfaces that once absorbed rainwater. When these surfaces are replaced with concrete, rainwater has nowhere to go, overwhelming drainage systems and causing waterlogging.

The reintroduction of green spaces—such as urban forests, roadside plantations, and bioswales—can improve the city’s water absorption and retention capacity. Green areas allow for groundwater recharge and delay surface runoff, functioning as a buffer during heavy rains. Rooftop gardens also play a critical role in rainwater harvesting and slowing down the flow of water into drains. In a city struggling with both drought and flood cycles, such solutions offer vital dual benefits.

10.4 Health and Wellness Benefits

Access to green environments is strongly correlated with better mental and physical health outcomes. In Chennai, where many neighborhoods lack public parks or gardens, the stress of daily commuting, overcrowding, and pollution takes a toll on residents’ health. Increasing green spaces can offer tangible psychological relief and foster a more active lifestyle.

Residents living near green areas tend to engage more in walking, jogging, and outdoor recreation, leading to improved cardiovascular health and lower obesity rates. Additionally, even passive interaction—like viewing greenery through a window—has been associated with reduced anxiety, improved concentration, and better mood regulation. Hospitals and schools surrounded by greenery have recorded faster patient recovery rates and enhanced student performance, respectively. For the aging population, shaded paths and accessible parks provide safe environments for physical activity and social interaction, thereby reducing loneliness and improving mobility.

10.5 Social Cohesion and Aesthetic Value

Beyond health and climate benefits, greenery improves the social and cultural fabric of a city. Shared green spaces encourage public gatherings, community events, and daily interactions, fostering a stronger sense of neighborhood identity and

cooperation. Initiatives such as community gardens and citizen-maintained parks empower residents and promote civic responsibility.

From a visual perspective, green infrastructure breaks the monotony of grey urban landscapes, adding color, texture, and seasonal variation. This enhances the city's visual appeal, attracts tourism, and can even influence the economic value of surrounding real estate. Areas with attractive greenery often experience higher foot traffic, increased local business activity, and reduced vandalism and crime.

Moreover, children who grow up with access to nature are more likely to develop environmental consciousness, creativity, and emotional resilience. The aesthetic and social dimensions of greenery thus contribute to building a healthier, happier, and more cohesive urban society.

11. Conclusion

This study has explored the complex relationship between urban development and greenery through a comparative lens, focusing on two rapidly evolving metropolitan areas: Chennai, India, and Shanghai, China. Drawing on two decades of NDVI-based satellite data, contributing environmental and socio-economic variables, and predictive modeling, the findings illustrate the stark contrast in ecological planning and its long-term outcomes.

11.1 Summary of Findings

The historical analysis from 2005 to 2024 reveals a steady decline in green cover across Chennai, largely due to unregulated urban sprawl, ineffective land-use policies, and insufficient prioritization of environmental buffers. NDVI values consistently fell year-over-year, especially in high-density zones that experienced infrastructural growth without corresponding investments in greenery. Correlations with increasing temperature, rainfall variability, and pollution underscore the role that declining green spaces have played in worsening the city's livability.

In contrast, Shanghai has shown a steady increase in NDVI values over the same period. This trend aligns with deliberate government intervention, structured afforestation campaigns, urban reforestation zones, and incentives for rooftop greening. Notably, Shanghai's growth has not come at the cost of environmental degradation; instead, the city has used its economic advancement as a platform to implement ecological enhancements. This dual growth—both vertical (infrastructure) and horizontal (green expansion)—highlights the possibility of integrated urban planning.

11.2 Importance of Predictive Modeling in Urban Planning

One of the critical contributions of this study is the use of predictive modeling to forecast NDVI trends for the next 20 years. These models serve as both analytical tools and policy instruments. They allow planners and decision-makers to see beyond the immediate present and visualize the consequences of current trajectories. For Chennai, the predicted projections from 2025 to 2044 show a further decline unless immediate and effective interventions are implemented. In contrast, Shanghai's models forecast continued ecological growth if current policies are maintained.

These simulations go beyond mere prediction—they can be used to model hypothetical interventions, simulate the effect of new greening laws, and optimize

land use to balance urban needs with environmental health. The integration of machine learning techniques like LSTM and multivariate regression enhances forecasting accuracy and contextual relevance, turning data into actionable insight.

11.3 Urgency for Green Reform in Chennai

Chennai finds itself at a critical crossroads. The past two decades of unchecked urbanization have not only reduced its green buffer zones but have also exposed the city to more extreme climate events—heatwaves, urban flooding, and declining air quality. Left unaddressed, these issues will compound, putting at risk the city’s economic vitality and public health.

The case of Shanghai demonstrates that green reform is not just environmentally desirable, but also economically and socially strategic. Chennai must now prioritize environmental recovery as a central aspect of its urban policy. Delaying action will not only make future interventions costlier but will also jeopardize the well-being of millions.

11.4 Vision for Sustainable City Development

This study ultimately envisions a Chennai where growth and greenery coexist—a city where rooftops bloom, vertical walls host gardens, and streets are shaded by thriving tree canopies. Such a transformation demands integrated planning across departments, strong political will, and citizen participation. From zoning reforms that mandate minimum green space per construction project to community-based reforestation and rooftop garden subsidies, the roadmap is both feasible and impactful.

A sustainable Chennai is not just about aesthetics; it is about creating a resilient, breathable, and livable city that safeguards its natural assets while embracing modernization. The tools, data, and global precedents already exist—what remains is for policymakers, planners, and citizens to act with urgency and unity.

12. Appendix

Code snippets:

```
import ee
import pandas as pd
import os

# Authenticate and initialize Earth Engine
ee.Authenticate()
ee.Initialize()

# Define Shanghai bounding box
shanghai = ee.Geometry.Rectangle([120.85, 30.7, 122.2, 31.9])

# NDVI ImageCollection
ndvi_collection = ee.ImageCollection("MODIS/061/MOD13Q1").select("NDVI")

# Output path (set your desired path)
output_csv_path = "C:/maps/datasets/ndvi_shanghai_2004_2024.csv"

# List of years
years = list(range(2004, 2025))
ndvi_data = []

# Loop through each year and extract mean NDVI
for year in years:
    start_date = f"{year}-01-01"
    end_date = f"{year}-12-31"

    yearly_ndvi = ndvi_collection.filterDate(start_date, end_date).mean()
    stats = yearly_ndvi.reduceRegion(
        reducer=ee.Reducer.mean(),
        geometry=shanghai,
        scale=250,
        maxPixels=1e9
    )

    mean_ndvi = stats.get("NDVI").getInfo()
    if mean_ndvi is not None:
        ndvi_data.append((year, mean_ndvi / 10000)) # MODIS NDVI scale is *10000
    else:
        ndvi_data.append((year, None))

# Convert to DataFrame
df = pd.DataFrame(ndvi_data, columns=["Year", "NDVI"])

# Save to CSV
df.to_csv(output_csv_path, index=False)
print(f"✅ NDVI saved to: {output_csv_path}")
```

```

import ee
import pandas as pd

# Initialize Earth Engine
ee.Initialize()

# Define Shanghai geometry
shanghai = ee.Geometry.Polygon([
    [120.85, 30.68],
    [120.85, 31.90],
    [122.15, 31.90],
    [122.15, 30.68],
    [120.85, 30.68]
])

# Define years
years = list(range(2004, 2025))

# Function to calculate annual rainfall
def get_annual_rainfall(year):
    start_date = f'{year}-01-01'
    end_date = f'{year}-12-31'
    dataset = ee.ImageCollection('UCSB-CHG/CHIRPS/DAILY').filterDate(start_date, end_date).filterBounds(shanghai)
    annual_rainfall = dataset.select('precipitation').sum()
    mean_rainfall = annual_rainfall.reduceRegion(
        reducer=ee.Reducer.mean(),
        geometry=shanghai,
        scale=5000,
        maxPixels=1e13
    ).get('precipitation')
    return {'year': year, 'rainfall': mean_rainfall.getInfo()}

# Compute rainfall for each year
rainfall_data = [get_annual_rainfall(year) for year in years]

# Convert to DataFrame
df_rainfall = pd.DataFrame(rainfall_data)

# Save to CSV
output_path = 'C:/maps/datasets/shanghai_rainfall_2004_2024.csv'
df_rainfall.to_csv(output_path, index=False)

```

```

# Function to calculate annual temperature
def get_annual_temperature(year):
    start_date = f'{year}-01-01'
    end_date = f'{year}-12-31'
    dataset = ee.ImageCollection('ECMWF/ERA5_LAND/DAILY_AGGR').filterDate(start_date, end_date).filterBounds(shanghai)
    annual_temp = dataset.select('temperature_2m').mean()
    mean_temp = annual_temp.reduceRegion(
        reducer=ee.Reducer.mean(),
        geometry=shanghai,
        scale=1000,
        maxPixels=1e13
    ).get('temperature_2m')
    return {'year': year, 'temperature': mean_temp.getInfo() - 273.15} # Convert from Kelvin to Celsius

# Compute temperature for each year
temperature_data = [get_annual_temperature(year) for year in years]

# Convert to DataFrame
df_temperature = pd.DataFrame(temperature_data)

# Save to CSV
output_path = 'C:/maps/datasets/shanghai_temperature_2004_2024.csv'
df_temperature.to_csv(output_path, index=False)
print(f'Temperature data saved to {output_path}')

```

```

import ee
import pandas as pd

ee.Initialize()

shanghai_geometry = ee.Geometry.Rectangle([120.85, 30.68, 122.12, 31.88])

available_years = [2000, 2005, 2010, 2015, 2020]

def get_population(year):
    dataset_id = f'WorldPop/GP/100m/pop/CHN_{year}'
    try:
        image = ee.Image(dataset_id)
        population = image.reduceRegion(
            reducer=ee.Reducer.sum(),
            geometry=shanghai_geometry,
            scale=100,
            maxPixels=1e12
        )
        total = population.get('population').getInfo()
        return {'year': year, 'population': total}
    except Exception as e:
        print(f"Error for year {year}: {e}")
        return {'year': year, 'population': None}

population_data = [get_population(year) for year in available_years]

df = pd.DataFrame(population_data)
df.to_csv('C:/maps/datasets/shanghai_population.csv', index=False)
print("Saved to C:/maps/datasets/shanghai_population.csv")

```

```

import pandas as pd

# Manually enter available population data for Shanghai (example values, replace with actual)
data = {
    'year': [2000, 2005, 2010, 2015, 2020],
    'population': [17000000, 18500000, 20000000, 23000000, 24800000] # Replace with actual values
}

df = pd.DataFrame(data)

# Interpolate yearly data between 2000 and 2020
df_full = df.set_index('year').reindex(range(2000, 2021)).interpolate(method='linear').reset_index()
df_full.columns = ['year', 'population']

# Save to CSV
df_full.to_csv('C:/maps/datasets/shanghai_population_2000_2020_interpolated.csv', index=False)
print("Interpolated yearly population saved.")

```

```

import ee
import pandas as pd

ee.Initialize()

shanghai_geometry = ee.Geometry.Rectangle([120.85, 30.68, 122.12, 31.88])

years = list(range(2004, 2020))

def get_urban_area(year):
    try:
        dataset = ee.Image(f'MODIS/006/MCD12Q1/{year}_01_01')
        land_cover = dataset.select('LC_Type1')
        urban_mask = land_cover.eq(13)
        urban_pixels = urban_mask.reduceRegion(
            reducer=ee.Reducer.sum(),
            geometry=shanghai_geometry,
            scale=500,
            maxPixels=1e13
        ).get('LC_Type1')
        total_pixels = land_cover.reduceRegion(
            reducer=ee.Reducer.count(),
            geometry=shanghai_geometry,
            scale=500,
            maxPixels=1e13
        ).get('LC_Type1')
        urban_percent = ee.Number(urban_pixels).divide(ee.Number(total_pixels)).multiply(100).getInfo()
        return {'year': year, 'urban_percent': urban_percent}
    except Exception as e:
        print(f"Urban error {year}: {e}")
        return {'year': year, 'urban_percent': None}

urban_data = [get_urban_area(year) for year in years]
df_urban = pd.DataFrame(urban_data)
df_urban.to_csv('C:/maps/datasets/shanghai_urban_2004_2019.csv', index=False)
print("Saved: C:/maps/datasets/shanghai_urban_2004_2019.csv")

```

```

def get_no2_for_year(year):
    try:
        start = f'{year}-01-01'
        end = f'{year}-12-31'

        no2 = ee.ImageCollection('COPERNICUS/S5P/OFFL/L3_NO2') \
            .filterDate(start, end) \
            .select('NO2_column_number_density') \
            .mean()

        mean_dict = no2.reduceRegion(
            reducer=ee.Reducer.mean(),
            geometry=shanghai_geometry,
            scale=1000,
            maxPixels=1e12
        )

        mean_no2 = mean_dict.get('NO2_column_number_density').getInfo()
        return {'year': year, 'no2_mean': mean_no2}
    except Exception as e:
        print(f"NO2 error {year}: {e}")
        return {'year': year, 'no2_mean': None}

no2_years = list(range(2018, 2025))
no2_data = [get_no2_for_year(year) for year in no2_years]
df_no2 = pd.DataFrame(no2_data)
df_no2.to_csv('C:/maps/datasets/shanghai_no2_2018_2024.csv', index=False)
print("Saved NO2 data to C:/maps/datasets/shanghai_no2_2018_2024.csv")

```

```

import pandas as pd
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

# Load combined dataset
data = pd.read_csv("C:/maps/datasets/shanghai_combined_2004_2024.csv")
years = data['year']
X = data.drop(columns=['year']).values.T
# Set number of clusters
n_clusters = 3

# Apply Fuzzy C-Means clustering
cntr, u, u0, d, jm, p, fpc = fuzz.cluster.cmeans(
    X, c=n_clusters, m=2, error=0.005, maxiter=1000, init=None)
labels = np.argmax(u, axis=0)
data['cluster'] = labels
data['year'] = years

data.to_csv("C:/maps/datasets/shanghai_fcm_clustered_2004_2024.csv", index=False)
print("Clustering done and saved to shanghai_fcm_clustered_2004_2024.csv")

```

```

import pandas as pd
import matplotlib.pyplot as plt
from prophet import Prophet
import os
file_path = r'C:\maps\datasets\shanghai_fcm_clustered_2004_2024.csv'
df = pd.read_csv(file_path)
df.columns = df.columns.str.strip().str.lower()
df.rename(columns={'year': 'ds', 'ndvi': 'y'}, inplace=True)
df['ds'] = pd.to_datetime(df['ds'], format='%Y')
model = Prophet()
model.fit(df)
future = model.make_future_dataframe(periods=20, freq='Y')
forecast = model.predict(future)
predicted = forecast[forecast['ds'].dt.year > 2024].copy()
predicted_dataset = predicted[['ds', 'yhat', 'yhat_lower', 'yhat_upper']].copy()
predicted_dataset['Year'] = predicted_dataset['ds'].dt.year
predicted_dataset.rename(columns={
    'yhat': 'Predicted_NDVI',
    'yhat_lower': 'Lower_Bound',
    'yhat_upper': 'Upper_Bound'
}, inplace=True)
predicted_dataset = predicted_dataset[['Year', 'Predicted_NDVI', 'Lower_Bound', 'Upper_Bound']]
save_path = r'C:\maps\datasets\ndvi_predictions_2025_2044.csv'
os.makedirs(os.path.dirname(save_path), exist_ok=True)
predicted_dataset.to_csv(save_path, index=False)
print(f"\n✅ Predicted NDVI values saved to: {save_path}")
print("\n🔍 Preview:")
print(predicted_dataset.head())

```

```

import cv2
import numpy as np
import os
image_path = "C:/maps/chennaiimap.png"
base_image = cv2.imread(image_path)
if base_image is None:
    raise FileNotFoundError("Image not found at the specified path.")
hsv = cv2.cvtColor(base_image, cv2.COLOR_BGR2HSV)
lower_green = np.array([35, 40, 40])
upper_green = np.array([85, 255, 255])
green_mask = cv2.inRange(hsv, lower_green, upper_green)
contours, _ = cv2.findContours(green_mask, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)
ndvi_values = [
    0.0840, 0.0746, 0.0717, 0.0676, 0.0648, 0.0623, 0.0602, 0.0585, 0.0571, 0.0559,
    0.0549, 0.0541, 0.0534, 0.0529, 0.0524, 0.0520, 0.0517, 0.0514, 0.0512, 0.0509
]
output_dir = "C:/maps/generated_maps"
os.makedirs(output_dir, exist_ok=True)
for i, ndvi in enumerate(ndvi_values):
    adjusted_hsv = hsv.copy()

    factor = ndvi / max(ndvi_values)
    adjusted_hsv[:, :, 1] = np.where(green_mask > 0, adjusted_hsv[:, :, 1] * factor, adjusted_hsv[:, :, 1])

    removal_fraction = ((max(ndvi_values) - ndvi) / (max(ndvi_values) - min(ndvi_values))) * 0.5
    num_contours_to_shrink = int(len(contours) * removal_fraction)
    if num_contours_to_shrink > 0:
        for j in range(num_contours_to_shrink):
            if j < len(contours):
                contour_mask = np.zeros_like(green_mask)
                cv2.drawContours(contour_mask, contours, j, (255), thickness=cv2.FILLED)
                kernel_size = 7
                blurred_hsv = cv2.GaussianBlur(adjusted_hsv, (kernel_size, kernel_size), 0)

                adjusted_hsv = np.where(contour_mask[..., None] > 0, blurred_hsv, adjusted_hsv)
    adjusted_image = cv2.cvtColor(adjusted_hsv, cv2.COLOR_HSV2BGR)

    output_path = os.path.join(output_dir, f"ndvi_map_{i+1}.png")
    cv2.imwrite(output_path, adjusted_image)
    print(f"Saved: {output_path}")

```

```

import cv2
import numpy as np
import os
import pandas as pd
ndvi_df = pd.read_csv("C:/maps/datasets/ndvi_predictions_2025_2044.csv")
ndvi_values = ndvi_df["Predicted_NDVI"].tolist()
ndvi_values.sort()
image_path = "C:/maps/shanghaimap.png"
base_image = cv2.imread(image_path)
if base_image is None:
    raise FileNotFoundError("Image not found at the specified path.")
hsv = cv2.cvtColor(base_image, cv2.COLOR_BGR2HSV)
lower_green = np.array([35, 40, 40])
upper_green = np.array([85, 255, 255])
green_mask = cv2.inRange(hsv, lower_green, upper_green)

output_dir = "C:/maps/generated_maps2"
os.makedirs(output_dir, exist_ok=True)

max_ndvi = max(ndvi_values)

for i, ndvi in enumerate(ndvi_values):
    adjusted_hsv = hsv.copy()

    sat_factor = (ndvi / max_ndvi) * 1.6
    val_factor = (ndvi / max_ndvi) * 1.3

    adjusted_hsv[:, :, 1] = np.where(
        green_mask > 0,
        np.clip(adjusted_hsv[:, :, 1] * sat_factor, 0, 255),
        adjusted_hsv[:, :, 1]
    )
    adjusted_hsv[:, :, 2] = np.where(
        green_mask > 0,
        np.clip(adjusted_hsv[:, :, 2] * val_factor, 0, 255),
        adjusted_hsv[:, :, 2]
    )

    final_image = cv2.cvtColor(adjusted_hsv, cv2.COLOR_HSV2BGR)

    out_path = os.path.join(output_dir, f"shanghai_ndvi_map_{i+1}.png")
    cv2.imwrite(out_path, final_image)
    print(f"Saved: {out_path}")

print("✅ All maps generated with smooth, natural increase in greenery 🌿")

```

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